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MASTER'S THESIS

"Interpreting deep neural networks: Explaining the methods behind AI-generated image recognition"

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Abstract

Generative Adversarial Networks (GANs), particularly StyleGAN, have achieved remarkable success in synthesizing high-fidelity images, yet their internal mechanisms, especially those of the discriminator component, often remain opaque.

The deficiency of transparency hampers several key aspects, including model validation, debugging, trustworthiness, and the ability to mitigate biases. This thesis addresses the challenge of interpreting the StyleGAN discriminator for high-resolution facial image generation, focusing on the FFHQ dataset. A discriminator network was trained from scratch against a fixed, pre-trained StyleGAN generator. The core methodology involved applying gradient-based interpretability techniques, namely Grad-CAM and Gradient Saliency Maps, to visualize the features and image regions that influence the decisions the discriminator makes when classifying real versus synthetic images. Performance was assessed with standard metrics (accuracy, precision, recall, F1-score, and AUC-ROC) to measure how well the discriminator was classifying the images, while also allowing for a bit of qualitative analysis to peek inside the black box and see how the discriminator was making these classifications and to obtain a universal interpretation of its decision-making strategy. This work demonstrates the adequacy of using Grad-CAM and Saliency Maps for interpreting StyleGAN discriminators and how this connects to the overall work of analyzing and understanding GANs. Results showed that the StyleGAN discriminator exhibited very good performance. AUC = 1.00 and accuracy = 1.00, indicating that it was learning how to discriminate very well. Qualitative assessment uncovered that the discriminator depends on not only high-level semantic facial features (like the eyes, the nose, or the mouth) but also relies on fine-grained textural details that are highlighted in the visualizations. This research serves to validate the use of Grad-CAM and saliency maps for dissecting the behaviour of the StyleGAN discriminator, considering both the not-so-obvious parts and the obvious parts that the network has learned to use in telling real faces from generated ones. It provides a more reliable framework for analysing GANs.

Declaration of Authorship

I, Nikunjkumar Tandel, hereby declare that this master's thesis, entitled "**Interpreting deep neural networks: Explaining the methods behind AI-generated image recognition**" is my own work completed without the use of any aids other than those listed. Any material from other sources or works done by others has been given due acknowledgment and listed in the reference section. Quoted sentences are marked with quotation marks, and other references related to the statement and scope of the work are also included.

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Signature

Acknowledgements

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List of Acronyms and Abbreviations

- **AdaIN:** Adaptive Instance Normalization
- **ADA:** Adaptive Discriminator Augmentation
- **AE:** Autoencoder
- **AI:** Artificial Intelligence
- **ALI:** Adversarially Learned Inference
- **ANOVA:** Analysis of Variance
- **AUC:** Area Under the Curve (often referring to AUC-ROC)
- **BCE:** Binary Cross-Entropy
- **BiGAN:** Bidirectional Generative Adversarial Network
- **CAD:** Computer-Aided Detection/Design (confirm context)
- **CC:** Correlation Coefficient
- **cGAN:** conditional Generative Adversarial Network
- **CLIP:** Contrastive Language–Image Pre-training
- **CNN:** Convolutional Neural Network
- **Conv2d:** Two-Dimensional Convolution Layer
- **CT:** Computerized Tomography
- **DCGAN:** Deep Convolutional Generative Adversarial Network
- **DP:** Dual-Phase (Steel)
- **EMA:** Exponential Moving Average
- **EMD:** Earth Mover's Distance
- **FA:** Feature Augmentation
- **FFHQ:** Flickr-Faces-HQ (Dataset)
- **FID:** Fréchet Inception Distance
- **FN:** False Negative
- **FP:** False Positive
- **FPR:** False Positive Rate
- **FVBN:** Fully Visible Belief Network
- **G:** Generator (in GANs)
- **GAN:** Generative Adversarial Network
- **GPU:** Graphics Processing Unit
- **Grad-CAM:** Gradient-weighted Class Activation Mapping
- **HCI:** Human-Computer Interaction
- **InfoGAN:** Information Maximizing Generative Adversarial Network
- **IoU:** Intersection over Union
- **IP:** Information Pursuit
- **IS:** Inception Score
- **KID:** Kernel Inception Distance
- **KL:** Kullback-Leibler (Divergence)
- **LBP:** Local Binary Patterns
- **LeakyReLU:** Leaky Rectified Linear Unit
- **LIME:** Local Interpretable Model-agnostic Explanations
- **LPIPS:** Learned Perceptual Image Patch Similarity

- **LR:** Learning Rate
- **LSGAN:** Least Squares Generative Adversarial Network
- **mAP:** mean Average Precision
- **MI-GAN:** Medical Imaging Generative Adversarial Network
- **MLP:** Multi-Layer Perceptron
- **mIoU:** mean Intersection over Union
- **MRI:** Magnetic Resonance Imaging
- **MS-SSIM:** Multi-Scale Structural Similarity Index
- **MSG-GAN:** Multi-Scale Gradient Generative Adversarial Network
- **NLL:** Negative Log-Likelihood
- **NLP:** Natural Language Processing
- **NSS:** Normalized Scanpath Saliency
- **OASIS:** (Ordinal Adversarial Supervision for Image Synthesis - Confirm definition if used)
- **OSMIS:** One-Shot joint Image-Mask Synthesis
- **PCA:** Principal Component Analysis
- **PET:** Positron Emission Tomography
- **PPL:** Perceptual Path Length
- **ReLU:** Rectified Linear Unit
- **RGAN:** Relativistic Generative Adversarial Network
- **RMSprop:** Root Mean Square Propagation (Optimizer)
- **ROI:** Region of Interest
- **ROC:** Receiver Operating Characteristic
- **SEM:** Scanning Electron Microscope
- **SeqGAN:** Sequence Generative Adversarial Network
- **SGD:** Stochastic Gradient Descent
- **SHAP:** SHapley Additive exPlanations
- **SIS:** Semantic Image Synthesis
- **SIV-GAN:** Single Image/Video Generative Adversarial Network
- **SPADE:** Spatially Adaptive Normalization
- **SR:** Super-Resolution
- **SRGAN:** Super-Resolution Generative Adversarial Network
- **SSIM:** Structural Similarity Index
- **SSL:** Self-Supervised Learning
- **SVM:** Support Vector Machine
- **TIS:** Temporal Inception Score
- **TN:** True Negative
- **TNR:** True Negative Rate (Specificity)
- **TP:** True Positive
- **TPR:** True Positive Rate (Recall/Sensitivity)
- **VAE:** Variational Autoencoder
- **VDA:** Visual Domain Adaptation
- **VGG:** Visual Geometry Group (Network)
- **VQA:** Video Quality Assessment

- **W:** Intermediate Latent Space (in StyleGAN)
- **WGAN:** Wasserstein Generative Adversarial Network
- **WGAN-GP:** Wasserstein Generative Adversarial Network with Gradient Penalty
- **XAI:** Explainable Artificial Intelligence
- **Z:** Input Latent Space (in GANs)

1 Introduction

1.1 Background and Motivation

Deep learning has experienced a paradigm shift with the advent of Generative Adversarial Networks (GANs), a class of models introduced by Goodfellow et al.[1] that has fundamentally altered the landscape of image synthesis and manipulation. GANs have quickly become a core technology, allowing for the generation of strikingly realistic visual content. At the heart of a GAN lies a competitive interplay between two neural networks: a generator, which synthesizes novel data instances (like pictures), and a discriminator, which evaluates those instances and their authenticity, distinguishing them from real-world data. This process compels the generator to iteratively improve its output (now with the help of the Maths and Magic we have thing to call the 'Wasserstein Metric'), allowing for the generation of images that have pictures' qualities (Figure 1). The reach of GANs is broad, with applications spread across many different areas, including sophisticated image-to-image translation [2],[3],strikingly effective data augmentation strategies [4],[5], and creative style transfer [6].

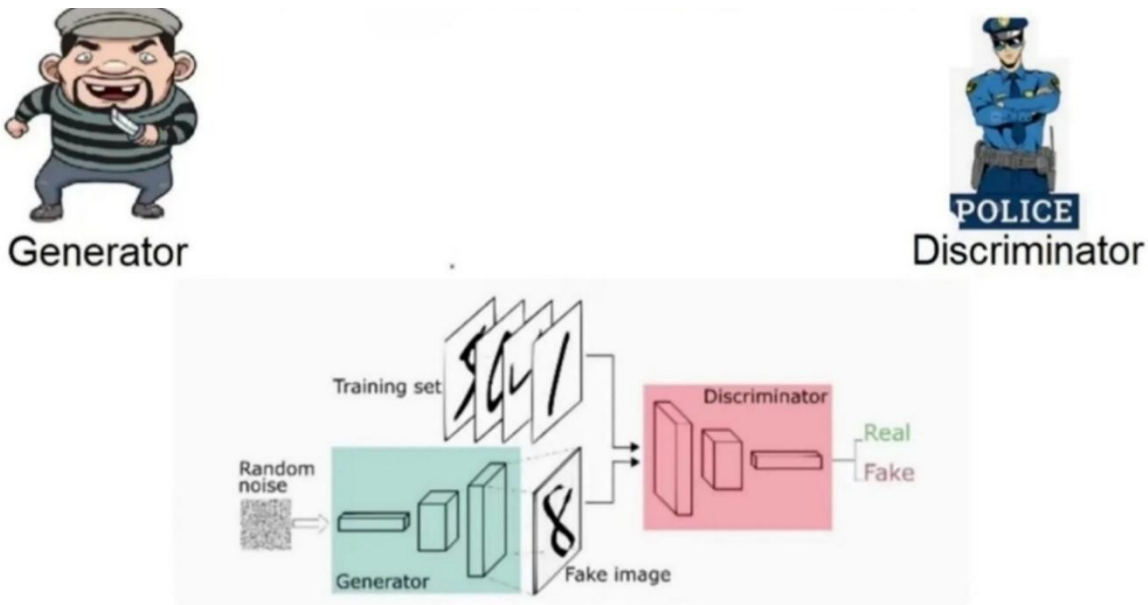

Figure 1 components of GANs [7]

This figure illustrates Conceptual diagram of the Generative Adversarial Network framework, illustrating the generator (G) producing synthetic data from noise (z) and the discriminator (D) classifying real and synthetic data, with feedback influencing both networks.

Within the expanding taxonomy of GAN architectures, StyleGAN [8] has emerged as a particularly influential model. It is distinguished by exceptional capacity for generating high-resolution, high-fidelity images coupled with sophisticated, fine-grained, control over stylistic elements [9],[10]. Unlike earlier GAN models, StyleGAN introduces truly innovative architectural features-most notably, a dedicated mapping network that transforms latent vectors into an intermediate style space (W space). And in the synthesis network, StyleGAN also uses

a more visually perplexing, but no less effective, way of controlling visual attributes called "adaptive instance normalization" (or $AdaIN$). Picture it this way: If the earlier GAN models were like a painter using a brush and palette, with StyleGAN, we now have an artist who is controlling the computer that does the painting. And it works at a truly staggering level of detail. Almost as good as a photograph.

Despite the striking visual quality achieved by models such as StyleGAN, there remains a big problem: the difficulty of understanding the internal decision-making of the discriminator network. Understanding the internal workings of the discriminator is not a mere academic fancy [11]; it is essential if we wish to improve the operation of GANs and ensure that they produce high-quality and trustworthy results. Discriminators are used to separate images that have been made by humans from those which have been "made" (not to say "drawn") by a GAN [12]. Certainly, for any humans who are looking at any images of the sort being "produced" by a GAN, it is far better if the images looked like the sort of thing you would find in the average stock photo portfolio than if they look like the sort of thing one finds in any collection of mediocre art. And yet the GAN gets no help from the human who is using the discriminator to make judgments about what is real and what is fake [13].

Advanced gradient-based visualization techniques have promising methodologies to penetrate this 'black box.' They have methods like Gradient Saliency Maps and Grad-CAM that help understand how deep neural networks including GAN discriminators work and create visual explanations that show which parts of an image are most important for the network's decisions. By highlighting influential image regions-those areas most critical to the network's classification output-these techniques give researchers insights into the discriminator's classification criteria. What visualizations like this can also do is reveal the specific features, textures, or patterns that the discriminator finds salient, which could help uncover potential biases learned from the training data. Complementary research, like that described in a pair of recent papers on StylEx, aims to explain classifier decisions within StyleGAN's StyleSpace [14].This work relates the low-level structure of images in StyleSpace with the high-level structure of images as seen by a classifier. When the structures in the two spaces relate in a certain way, it gives a certain unexpected result-unexpected to the classifier, that is, because it tries very hard to be good at its job. The result it tries to produce is a secret handshake between what the image looks like in StyleSpace and what it looks like at the high level of what was formerly called 'visual semantics.'

1.2 Research Problem and Objectives

This dissertation is about the StyleGAN discriminator. In particular, it is about the StyleGAN discriminator when it is faced with high-resolution facial images that have been rendered by the world-class StyleGAN generator [8]. These authentic and synthetic images together with their distinguishing features have been assembled into a dataset called FFHQ (Flickr-Faces-HQ). The task of the discriminator is to divide these two kinds of images, authentic and synthetic, into two distinct classes. The problem is difficult, however, partly owing to the

presence in the dataset of slight naturalistic variations (imperfections) that in authentic images perform the function of distinguishing features [15].

To resolve this challenge, several interconnected objectives guide this research. The first of these is to train and assess a StyleGAN discriminator network from the ground up, using a fixed pre-trained StyleGAN generator and a well-defined subset from FFHQ. (This is all carried out according to the program laid out in Chapter 3) The training is closely watched, and the performance of the resulting discriminator is gauged quite rigorously. This is done by relying on some good old-fashioned metrics-stuff like accuracy, precision, recall, F1-score, and AUC-ROC. These are all very standard as far as judging the performance of a classifier goes. But what makes them particularly relevant here is that they also allow varying shades of pass/fail performance to be painted. And painting these varying shades is very important. Because just as poor performance is a bad thing, good performance that isn't well-understood is likewise suspect. The next objective not only takes a step further into the business of understanding the model but also begins to employ some new methodologies that have only quite recently come onto the scene. What this next part does is apply two quite powerful techniques from the field of Explainable Artificial Intelligence (XAI) to the trained discriminator. These two techniques-the visualization methods Grad-CAM [16] (Chapter 4) and Gradient Saliency Maps [17] (Chapter 4) serve as quite advanced ways of peering into the decision-making process of the trained discriminator.

1.3 Contributions and Significance

This thesis conveys several contributions relevant to the understanding and application of generative adversarial networks, with a distinct focus on the interpretability and evaluation of the StyleGAN discriminator.

The first major contribution is the empirical application and validation of a comprehensive interpretability framework for the StyleGAN discriminator. This framework is systematically leveraged in the context of high-resolution face generation and is designed to shed light on the opaque decision-making processes of the discriminator. The components of the framework are Gradient Saliency Maps and Grad-CAM. The first of these, Gradient Saliency Maps, is a method that has proven effective in the illumining the opaque decision layer of neural networks. The Saliency Maps act as a window into the mind of our GAN discriminator.

Second, this thesis details a quantitative and qualitative analysis of a StyleGAN discriminator trained from scratch against a fixed, Pre trained generator. The setup allows for an unfavourable aspect of the discriminator to be assessed. In this setup, we can evaluate how and why the StyleGAN discriminator sees what it does, and we can also evaluate the high performance that it can achieve against a highly detailed target. This part of the thesis serves as a case study to complement the large-scale analyses [7], [18] and surveys [19].

Thirdly, the work provides insights into the roles different feature types play, with particular emphasis on the discriminator's sensitivity to both high-level semantic facial structures and fine-grained, high-frequency textural details. This is based on the combined interpretation of

Grad-CAM and Saliency Map visualizations, as presented in Chapter 5,This understanding has implications for GAN development and could potentially help efforts aimed at texture synthesis improvement or artifact reduction [10], [20].

This research is important for several reasons. It pushes forward the frontier of Explainable AI (XAI), in that it shows interpretability methods can be successfully applied to very sophisticated (and often said to be too complex for their own good) generative models [14], [21] like GANs. It has practical payoff, too. Interpretability enhances transparency, so the findings offer a path toward more robust and trustworthy GAN-based systems, which matters for applications where performance and ethical soundness must go hand in hand, like deepfake detection and bias analysis. The research may also affect future exploratory work with GANs and similar models [1].

1.4 Thesis Structure

The dissertation is arranged to yield a distinct and sensible advance throughout the examination of StyleGAN discriminator interpretability and assessment. It starts with basic concepts in Literature review chapter, where a thorough literature review is given, embracing fundamental subjects such as GAN fundamentals, StyleGAN breakthroughs, the importance of interpretability, pertinent methods of explainable AI (XAI), and the present conditions of GAN discriminator interpretability.

The empirical framework is detailed in Methodology chapter. It describes the experimental design, data preparation, model architectures, training procedures, and the interpretability framework. It also covers the evaluation metrics used and the statistical analysis done.

The research outcomes are presented in Result Chapter, showcasing not only the key quantitative performance metrics but also some of the more salient qualitative findings that emerged from the use of the XAI techniques. With these results in mind, Discussion Chapter moves into a discussion and interpretation of the findings. This chapter sees the research results set in the context of existing literature, with some significant connections being made and discussed in terms of their practical implications.

The thesis ends with Conclusion Chapter, which recaps the main discoveries, contributions, shortcomings, and prospects of the research. Also included are supplementary materials in Appendices A-F, such as results of exploratory experiments, sample generator outputs, detailed hyperparameters, additional visualizations, and the nitty-gritty of the computational environment. This all adds up to a quite thorough accounting of the research done.

2 Literature Review

This chapter presents the basic ideas and relevant previous work that supports this thesis. It starts with an overview of GANs, their training dynamics, and associated problems. It then goes on to the details of the StyleGAN architecture, its innovations, and applications. Following that, the chapter gets into the importance and methods of interpretability in deep neural networks, focusing on GAN discriminators. Finally, it presents the state-of-the-art in GAN discriminator interpretability and contextualizes the contributions of this research.

2.1 Theoretical Foundations of GANs

2.1.1 Introduction to Generative Adversarial Networks

One significant paradigm in generative modelling is represented by Generative Adversarial Networks (GANs), which were introduced by Goodfellow and his colleagues [1]. GANs come down to an adversarial training process between two neural networks: a generator (G) and a discriminator (D). The generator synthesizes data samples (e.g., images) that should be statistically indistinguishable from any given data distribution, while the discriminator learns to tell apart the real data instances from the fake ones that the generator produces. This serves as a competitive dynamic and is often formulated as a minimax game [22]. The GAN framework, which has garnered much attention and achieved many impressive results, also has many variants and extensions.

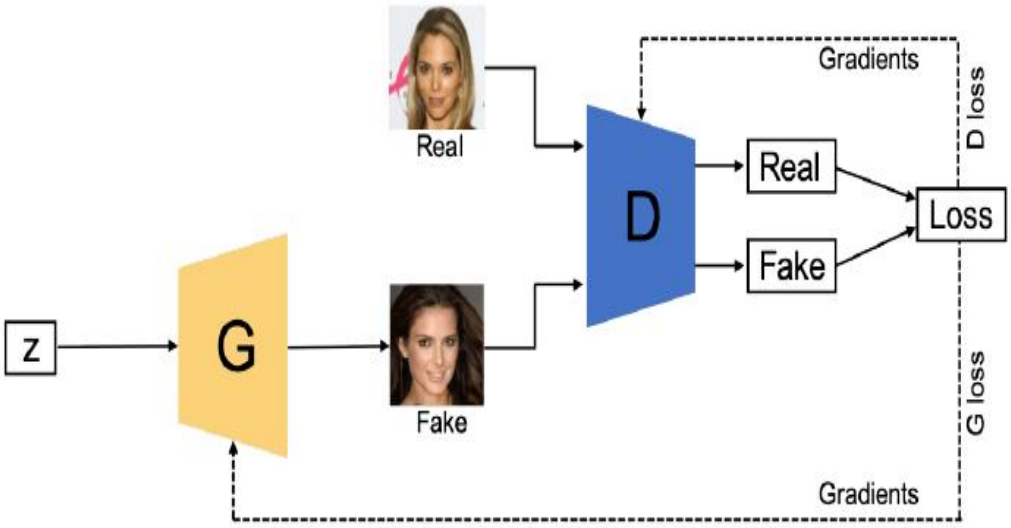

Figure 2 Conceptual diagram of the Generative Adversarial Network framework [23]

This figure illustrates the framework of the Generative Adversarial Network, providing a conceptual diagram in which the generator (G) synthesizes data from noise (z) and the discriminator (D) classifies the data as real or synthetic, all with feedback that influences both networks.

A stable equilibrium between the generator and discriminator is challenging to achieve in GAN training [24]. If the discriminator becomes too good relative to the generator, it can give the generator insufficient gradient feedback to learn effectively. When this happens, the generator itself may start to learn to produce only those samples to which the current top discriminator can only feebly respond. This danger of hitting a wall of too-good discriminators has been somewhat addressed in recent years by architectural diversity and advances in understanding just what the optimal discriminator might look like (or what families it might belong to). For example, an architecture choice with major implications for latent spaces is whether to use a single global discriminator or several local discriminators in a multi-component setup [25].

GANs have a remarkably diverse range of applications. They are used for image-to-image translation [2], [3] for data augmentation [4], [5] for style transfer for anomaly detection, and more. Within this landscape, the StyleGAN architecture has emerged as a particularly important model, renowned for its ability to synthesize very high-resolution images, very high-fidelity images, and especially human faces [19], [26] So, this thesis focuses on the StyleGAN discriminator, but not exclusively on it. It looks at the role of the StyleGAN discriminator in the very high-fidelity synthesis of images, and then it weirdly transitions into the topic of interpretability.

2.1.2 GAN Training Dynamics

It is crucial to understand the training dynamics of GANs for their further development and for the troubleshooting of resultant models. Training is itself a kind of minimax game between two neural nets: G and D [1].The generator learns to map an essentially arbitrary latent space (usually just noise, denoted as z) to the data space. Its goal is to produce samples that are indistinguishable from real data (i.e., it produces G(z) such that $p_data(x) \approx p_G(x)$). Meanwhile, the discriminator learns to tell real data apart from data generated by the current version of G. Its output is the probability that a given sample x is real (rather than generated).

The canonical GAN objective function expresses this adversarial relationship. The objective of GAN (Generative Adversarial Network) training is to find a G that minimizes the value of V while simultaneously finding a D that maximizes this value. Formally, we can express this:

$$\min_{G} \max_{D} V\,(D,G) E_{x\sim p_{\mathrm{d}}(x)}[logD(x)] + E_{z\sim p_z(z)}\left[log\left(1 - D\big(G(z)\big)\right)\right] \qquad \textit{Eq(2.1) [1]}$$

where

- $D(x)$Is the discriminator's probability that is real.
- $G(z)$ Is the generator output when fed noise Z.
- $P_d(x)$ Represents the real data distribution.
- $P_z(z)$ Is the random noise distribution used to generate synthetic data.

This equation represents the minimax objective. The discriminator D tries to maximize this value by correctly classifying real data (D(x) close to 1) and fake data ($D\big(G(z)\big)$ close to 0). The generator G tries to minimize this value by producing fake data G(z) that the discriminator classifies as real ($D\big(G(z)\big)$ close to 1). denotes expectation, $p_d(x)$ is the real data distribution, and $p_z(z)$ is the noise distribution.

The iterative process is ideally directed toward a Nash equilibrium, where the generator captures the real data distribution, and the discriminator outputs a probability of 0.5 for all inputs [27], [28] However, achieving this equilibrium in practice is very hard to do. As has been mentioned, and as we shall elaborate in Section 2, imbalances whereby D becomes too strong can lead to G gradients that vanish, while a weak D provides G with very poor guidance Achieving a proper balance in the iterative process towards the desired equilibrium state is a real problem in practice. Vanishing G gradient problems seem to be the most serious consequence of D's becoming either too strong or too weak.

Training stability has many possible roads leading to it. One is to modify the loss function directly, and there we have different Flavors of Earth Movers' Distance (also known as Wasserstein Distance). Our workhorse is probably the WGAN-GP (by Accomplice, aka NVIDIA) [29]. WGAN-GP gives very good gradients, and if D can use the very good gradients that WGAN-GP gives, then we can say that W is stable and (hopefully) D will not "overfit" too quickly. Now, WGAN-GP can work with different D architectures, and that gives us some freedom. In this part, we also have some regularization from using spectral normalization [19] and we also know (or at least we should know) that too deep of a D will lead to instability and have some features corresponding to the caveat that we can have high representational power but low generalization power [30].

2.1.3 Challenges in GAN Training

Effectively training GANs requires one to deal with several inherent challenges that stem from the adversarial framework. One main problem is instability in training, which often shows up as a phenomenon called mode collapse. Mode collapse occurs when the generator learns to produce only a few types of outputs that successfully fool the discriminator-it fails to capture the full variety that the training data comprises [31]. There's a reason that this is a problem: It stems from the difficulty in maintaining the delicate equilibrium between G and D. When D becomes too good, the amount of gradient information that it sends back to G becomes too small, and G learns to do different things but not to do any one thing well [24]. On the other hand, when D and G have converged to a state of equilibrium, if the next training iteration is not properly coordinated between G and D using well-thought-out gradient directions, the training will not converge. Visualization tools, like those used in this thesis (Grad-CAM, Saliency Maps), can potentially help in diagnosing such problems by revealing just which side (G or D) the instability is currently residing with.

Creating high-resolution images presents particular problems in trying to forge something that is both coherent and clear at the global structural level and the fine-grained, detailed, and local

level [8]. It is just too easy to generate clear pictures that include artifacts (and that are thus clear at the global level but not at the local level) or that are not consistent in terms of appearance or are just plain inconsistent - clear but with no detail [32].

GANs are still hard to evaluate. Usual metrics like Inception Score (IS) [33] or Fréchet Inception Distance (FID) [34] give some sort of a number, but they don't really capture the essence of what makes the pictures good or bad. They're just not that perceptually relevant. In fact, I consider them the kinds of evaluation that might be done by a grading robot. On a richer scale, the pictures' fidelity and diversity can also be measured into two separate halves using the generative model versions of precision and recall [35], [36]. But these are still just kind of half-measures.

Certain applications of GANs add extra layers of difficulty. For example, conditional GANs have to ensure that the outputs adhere closely to the conditioning inputs of text descriptions and semantic maps [25]. Then there are temporally coherent applications, like video generation, which are supposed to add additional richness to the outputs. All of these are supposed to be worked out in ongoing research into new architectures, more powerful training algorithms, and better evaluation methods [37], [38].

2.2 StyleGAN and Its Innovations

2.2.1 Overview of StyleGAN

StyleGAN [8] and its successors [10], [39],constitute a significant leap forward in GANs, especially with regard to their being known for synthesizing very high-resolution, very high-quality images, especially of human faces. They were devised by Tero Karras and his associates at NVIDIA. StyleGAN and its successors achieved several major architectural breakthroughs that have improved the way GANs are controlled-both in terms of the "style" as well as the "content" of images that they generate.

An important feature is the generator architecture based on style. Instead of directly feeding the latent code z into the synthesis network, StyleGAN first passes the code through a nonlinear mapping network (typically a multi-layer perceptron or MLP) to produce an intermediate code W. This code W is much less entangled than z, meaning that changes in W correspond much more directly to changes in the appearance of the generated image. For example, if you change something in W that's equivalent to changing a hairstyle, the image will show an obvious difference without changing other semantically important features like identity.

The intermediate latent code w (a vector in W) is then used to modulate the synthesis process at different convolutional layers via Adaptive Instance Normalization (AdaIN). $AdaIN$ adjusts the feature map statistics (mean and variance) based on style information derived from w, allowing control over visual characteristics ranging from coarse features (pose, shape) to fine details (color scheme, textures) [8].

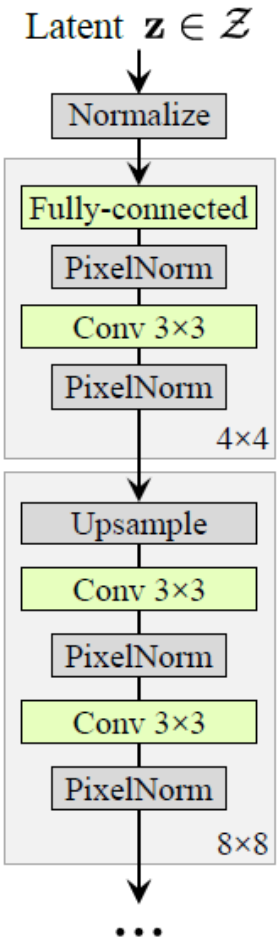

Figure 3 StyleGAN Generator Overview [8]

This figure presents a generator of the StyleGAN, showing how it takes a code vector z and reinterprets it into an intermediate code vector w, and then how it uses this vector and some noise to synthesize the final image. Adapted from Karras et al [8].

Stochastic variation is also incorporated into StyleGAN through direct noise inputs added to the feature maps at different resolutions. This enables the model to generate non-deterministic details (e.g., hair placement, freckles) that are independent from the main style information encoded in w. Further disentanglement of styles and improved image quality is achieved by a technique called style mixing regularization, where during training different w codes control different layer groups. Progressive growing was used by the original StyleGAN [9], but later versions (e.g., StyleGAN2) have achieved similar stability and high resolution through architectural refinements [10]. The discriminator in StyleGAN is also very important. It is designed to effectively evaluate the high-quality synthetic images and provide meaningful gradients back to the generator [37].

2.2.2 StyleGAN Architecture Details

Investigating the StyleGAN architecture in more detail reveals the reasons for its excellent performance [8], [10], The crucial transformation from the input latent space Z (for example,

512-dimensional Gaussian) to the intermediate latent space W (also often 512-dimensional) is performed by the mapping network, which is usually an 8-layer MLP. This non-linear mapping is critical for achieving the reduced feature entanglement compared to Z that is necessary for high-quality image synthesis.

The synthesis network builds the image progressively. It often starts from a learned constant input tensor rather than the latent code itself. At every convolutional block (working at the various resolutions of $4x4, 8x8, \ldots, 1024x1024)$, the intermediate latent code w is conditionally transformed to produce style parameters. These parameters are two: one is a coarsely determined scale (that performs horizontal adjustments across the feature maps); the other is a bias term (that makes fine-tuned vertical adjustments). Once modulated, the resulting feature maps are passed onto the next convolutional block (where another latent code conditionalization is performed), and this operation is repeated enough times.

$$\mathrm{AdaIN}(x_i, y) = y_{s,i}\frac{x_i - \mu(x_i)}{\sigma(x_i)} + y_{b,i} \qquad \textit{Eq(2.2) [40]}$$

This equation computes AdaIN first normalizes the i-th feature map x_i *(*$subtracting\ mean\ \mu, dividing\ by\ standard\ deviation\ \sigma$*), then scales and biases it is using style parameters* $y_{s,i}$*(scale) and* $y_{b,i}$*(bias) derived from the latent code w.*

which are derived from the latent code w. Furthermore, noise inputs sampled from a Gaussian distribution and scaled by learned per-channel factors are added to the feature maps within each block. This procedure allows the network to model stochastic variations like hair, freckles, or skin pores [8].

StyleGAN's models leverage a convolutional neural network (CNN) architecture with a high capacity for understanding visual representations. This means they can capture the kinds of details that impress and surprise us, like the shades of colour on an object's surface and the way light interacts with it, across multiple scales. The components of the CNN architecture-its layers, for instance-are well designed for this task. They are good at classifying image representations and, in a way, are also good at making us see much-improved representations of images that look even more human-like.

In addition, a combination of adversarial signals, regularization strategies, and architectural enhancements ensures that the generator is in control and remains consistent. This helps ensure that small perturbations in the latent space do not result in unrealistic or low-quality images. We believe that these design choices are significant in contributing to the model's ability to produce high-fidelity, photorealistic images that preserve both the semantic structure and graphical quality [10].

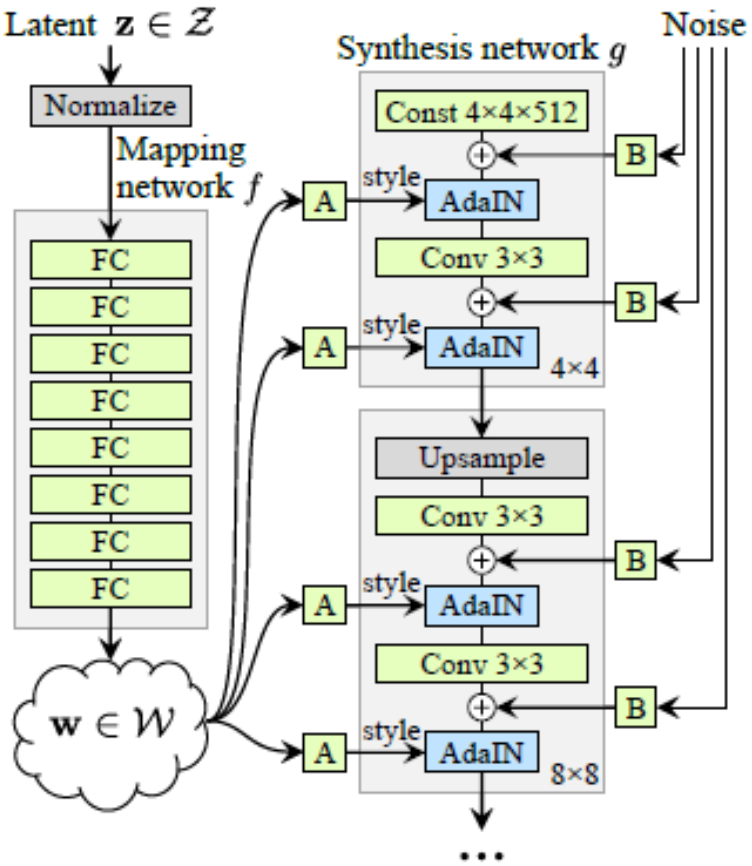

Figure 4 Detailed StyleGAN Generator Diagram[8]

This figure shows in greater detail a StyleGAN synthesis block, displaying the convolutional operation as well as the scaled noise addition and its place in the overall structure. It shows that AdaIN is applied with parameters derived from w, and we can see the fully convolutional layer that follows before the next upsampling happens.

2.2.3 StyleGAN Training Techniques & Applications

Training StyleGAN models effectively relies on specific techniques beyond the core architecture. One such technique is style mixing regularization [8] which is implemented as follows: for a percentage of training images, styles from two different latent codes, $w1$ and $w2$, are applied at different layer levels; coarse layers use $w1$, and fine layers use $w2$. This regularization is crucial for preventing the different layers of the model from learning correlated features, which would most likely happen without this component since the same features would be useful at these different levels. Correlated features hurt disentanglement, which we want to improve to make the model more interpretable. Path length regularization (PPL) [10], primarily used in the 2nd and later versions, helps a lot with this, ensuring that the mapping from W to the image space is well-behaved, leading to much better quality and editability of interpolated images. While not exclusive to StyleGAN, careful hyperparameter tuning (Appendix C(8.3)), appropriate loss functions, and potentially adaptive data augmentation strategies (like those used in StyleGAN-ADA) [39] are also vital GAN

stabilization techniques. (If you're wondering why the picture at the top of the page of the report isn't as good as the others, it's most likely because of several poor design choices made during the first few years of the project. Interpretability tools like Grad-CAM and saliency maps, while used post-hoc in this report, have potential for offering diagnostic insights during training [41].

Table 1 Key StyleGAN Applications

Application Area	**Description**
High-Fidelity Synthesis	Generation of highly realistic facial images, human bodies, and objects, often indistinguishable from real photos.
Semantic Image Editing	Allows manipulation of images by editing latent vectors—e.g., changing facial expressions, age, or pose.
Data Augmentation	Used to expand datasets for downstream tasks such as medical image classification or biometric training.
Representation Learning	Enables disentangled feature learning and supports studies in interpretability and generative factor control.
Deepfake Research	Used to understand the mechanics of deepfake generation and to aid in building robust detectors.
Bias Analysis	Helps expose and analyze biases in training data by observing outputs under different demographic conditions.

2.3 Interpretability in Deep Neural Networks

2.3.1 Importance of Interpretability in Style GAN

Understanding and explaining the internal mechanisms and decision-making processes of machine learning models, especially complex ones like deep neural networks, is what interpretability, or Explainable AI (XAI), is all about. The reasons for making the StyleGAN [42] discriminator interpretable are threefold. Firstly, if we wish to use the mechanism as a model validation and debugging tool, it must be understandable. If a human can understand why the discriminator would classify an image as real or fake, then we can more confidently assert that the discriminator is functioning well if it is relying on meaningful features that correlate with authenticity. If not, we might suspect it is paying too much attention to features that are more like a dataset artifact. Secondly, if GANs are eventually used in high-stakes applications (e.g., generating synthetic data for sensitive domains), then we should be able to trust the quality control mechanism that is the discriminator. If the discriminator is understandable, we can more confidently assert that its output is high quality, and that the system is trustworthy. Thirdly, and closely related to the previous two points, is that there are direct insights to be gained from an understandable discriminator that can be used to improve the GAN model itself. If we know where the mechanism is faltering, we can use that information to refine its constituent parts [34].

2.3.2 Gradient-Based Interpretability Methods

Gradient-based methods represent a prominent class of XAI techniques that leverage the gradient information flowing through the network to explain its predictions. These methods essentially attempt to figure out how changes in the input features affect the output score. One fundamental and probably the simplest technique is Gradient Saliency Maps [17]. As described in Chapter 3 (Section 3.4.2), this method consists of computing the gradient of the model's output score with respect to each input pixel. More precisely, it computes the derivative of the output score with respect to the inputs, pixel by pixel. The magnitude of this quantity, for each pixel, indicates how important that pixel was for the decision the model made.

Other gradient-based methods exist, such as Integrated Gradients [43], which address some theoretical limitations of simple gradients by integrating them along a path from a baseline input (e.g., a black image) to the actual input. Techniques like Guided Backpropagation [44] aim to produce sharper visualizations by modifying the backpropagation process, although their faithfulness to the model's actual computation has been questioned [36]. In the context of this thesis, Grad-CAM and basic (normalized) Gradient Saliency Maps were chosen as representative and widely used techniques for visualizing the StyleGAN discriminator's focus.

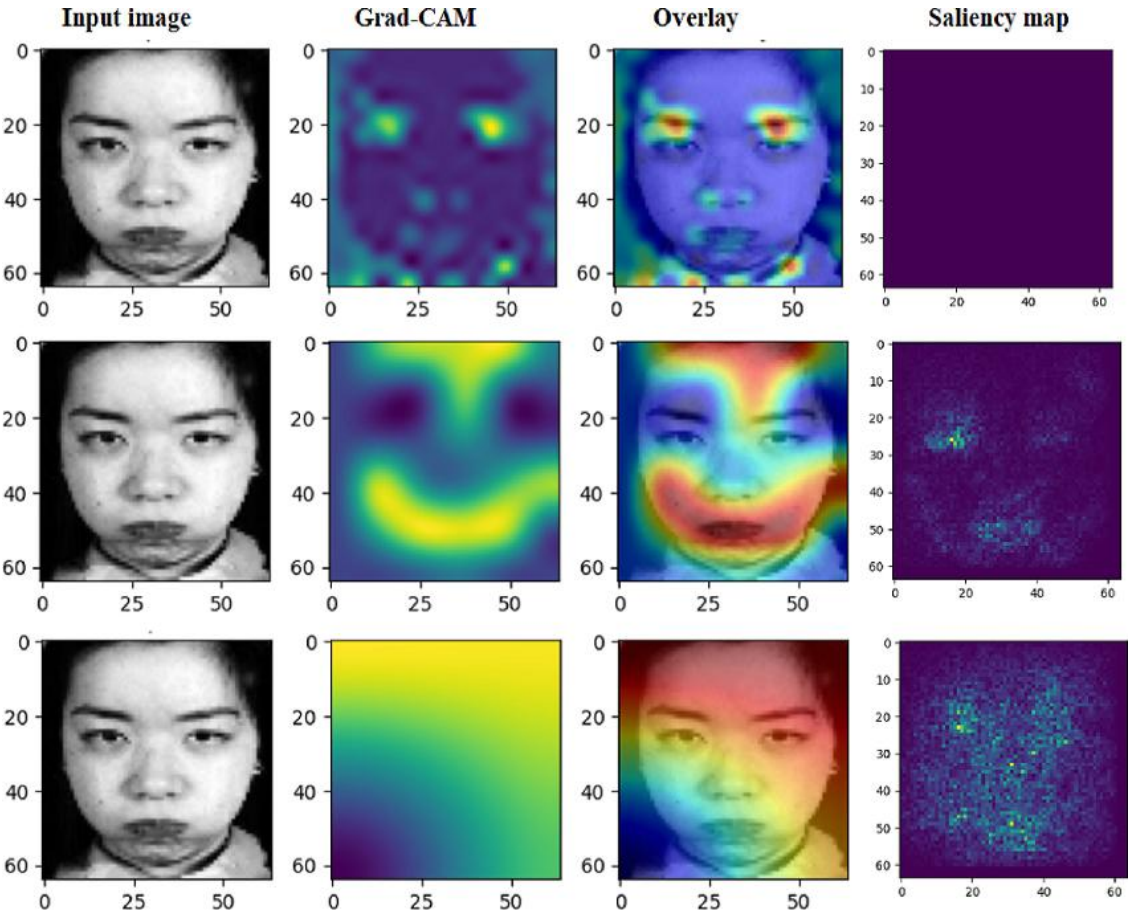

Figure 5 Comparison of Gradient Saliency Map and Grad-CAM [45]

This figure shows a conceptual comparison of interpretability methods based on gradients. Left: The input image. Centre: The heatmap from Grad-CAM that highlights which regions were influential. Right: The Gradient Saliency Map. It highlights which pixels were influential but does so at a very noisy level.

2.3.3 Activation-Based Interpretability Methods

Methods that are based on activation offer a complementary route to interpretability. They analyse the internal activations of neurons or feature maps within the network. This contrasts with the gradient-based methods we discussed earlier, which largely operate at a surface level and tell us how much of a change to expect in the output if we make a change to the input. Gradient-based methods tell us how to spell out the learned functions in terms of the inputs and the outputs. Activation-based methods tell us how the network is internalizing the features and representations in between the inputs and outputs.

The intermediate convolutional layers of a neural network can be directly inspected to see the patterns, textures, or shapes that certain filters within those layers have learned to recognize. When combined with knowledge of which filters activate most strongly for what inputs, these visualizations offer a window into the kind of reasoning that's going on inside the network at various levels of abstraction.

Activation maximization is a method that artificially creates an input image (beginning from random noise or a baseline image) that produces the greatest activation of a given neuron or layer. The image that results from this process serves as a visual representation of the preferred type of stimulus for that network component.

Techniques such as Network Dissection [18], [46] try to systematically associate the individual units (e.g., channels in a convolutional layer) of neural networks with understandable semantic concepts. They do this by measuring the alignment between unit activations and segmentation masks across a large dataset. Although these methods primarily apply to classifiers and generators, they could potentially adapt to analyse the units of discriminators as well. Activation Atlas [47] looks at the other end of the neural network: the decisions it makes. For a few images, it computes the activations that are produced by the network. Then, it uses those activations to visualize the space of learned features.

When it comes to the StyleGAN discriminator, studying activations in the last few convolutional layers just before the classification head might tell us something about the feature representations of real versus fake images. If the activations are indeed different, then maybe applying some sort of dimensionality reduction technique like PCA or t-SNE to the activation vectors might also reveal some sort of systematic difference between the representations of the different types of images or the different kinds of things that can go wrong when generating images with StyleGAN.

2.4 State of the Art in GAN Discriminator Interpretability

2.4.1 Existing Approaches

Interpretation of GAN discriminators for complex models like StyleGAN uses an ever-growing range of techniques. Central to these are visualization methods. As discussed, (Section 2.3.2) it is common to apply post-hoc methods like Gradient Saliency Maps and Grad-CAM to pinpoint the specific influential pixels or regions that lead to a certain decision. Of course, not all visualization acts to inform the user of the decision made for a particular input. Some act as the user's guide to the understand the reason why the model makes decisions in general. Activation-based visualization is a good example of a method that serves this purpose [18], [48], [49].

Exploring latent space is especially important for StyleGAN because it has an uncorrelated W space. This means that if you vary the W in a certain direction while keeping the other Ws constant, the variation will produce a type of change in the output that is relatively coherent and semantically meaningful. To put it another way, disentangled latent spaces make it easier to interpret the results of latent space exploration [50].

Modifications to the architecture aim to create discriminators that are more inherently interpretable. For instance, the authors of [51] suggest using segmentation-based outputs that provide real/fake feedback at the pixel level, while those in recommend using attention mechanisms to explicitly show which parts of an image are important.

Techniques of unit and probe analysis endeavour to comprehend the task of neurons or units in the discriminator's function. Network Dissection [18], [50],ambitiously attempts to associate units with semantic concepts and seeks to reveal whether the discriminator is learning meaningful representations of data or simply low-level artifacts.

Evaluating the Effect of the Discriminator on the Generator: Some studies treat the discriminator as a black box and try to infer its properties from the way it impacts the generator during training. They measure how well the generator uses the feedback from the discriminator to become a better generator. They also measure how well the generator can affect the discriminator. In these studies, it isn't clear whether to attribute the observed effects to the discriminator or to the way the generator uses the feedback [34].

The features pinpointed by interpretability methods are also validated through comparative analysis with human perception or expert knowledge to ensure that they indeed represent real, salient structures in the data, as opposed to interpretable artifacts [12]. The colours in Table 2 correspond to three different types of model interpretability approaches that were discussed previously.

Table 2 Comparison of GAN Discriminator Interpretability Approaches

Approach Category	**Description**	**Pros**	**Cons**
Gradient Visualization	Highlight input pixels/regions influencing output via gradients	Applicable post-hoc, Intuitive (Grad CAM)	Noisy (Saliency), Coarse (Grad CAM), Faithfulness?
Activation Visualization	Analyze internal neuron/feature map activations	Insights into learned representations	Less direct link to specific decisions
Latent Space Analysis	Probe sensitivity by manipulating generator latent codes	Links to semantic attributes (StyleGAN)	Indirect, Depends on generator quality
Architectural Design	Build inherently interpretable discriminators	Built-in interpretability	May constrain performance, Specific designs
Self-Supervised Tasks	Add auxiliary tasks to learn robust representations	Improves representation quality	Indirect interpretation

2.4.2 Comparative Analysis of Visualization Methods

Focusing on the primary visualization techniques used in this thesis, Gradient Saliency Maps and Grad-CAM a comparative analysis reveals complementary strengths and weaknesses for interpreting the StyleGAN discriminator. Gradient saliency maps offer pixel-level granularity, directly indicating the sensitivity of the discriminator's output score to infinitesimal changes in each input pixel. The results can be valuable for identifying fine-grained textural features or specific pixel-level artifacts that influence the decision [10]. However, their major drawback is susceptibility to visual noise and potentially highlighting non-robust features, which can make interpretation difficult [52],[36],[48]. Normalization helps, but careful interpretation is needed.

Grad-CAM, on the other hand, produces coarser heatmaps localized to broader image regions. By weighting feature maps from a convolutional layer based on class-specific gradients, it highlights areas deemed semantically important for the classification [16]. This form of classification often aligns better with human intuition about high-level features (like eyes and noses in facial images) and is generally less noisy than basic saliency maps [25]. However, its resolution is limited by the size of the target feature map, potentially missing fine-grained details captured by saliency methods. The choice of the target layer also influences the resulting visualization.

For analysing high-resolution StyleGAN/FFHQ images, Grad-CAM is often effective for understanding focus on major facial structures or identifying larger anomalous regions, while Saliency Maps (especially if noise is managed) might be better suited for pinpointing subtle textural differences or pixel-level artifacts [10]. There exists a trade-off between the faithfulness of gradient methods (how accurately they reflect the model's computation) and their interpretability (how easily humans can understand them) [44], [48]. Therefore, using both methods provides a more comprehensive view, correlating regional importance (Grad-CAM) with pixel-level sensitivity (Saliency Maps), as demonstrated by the combined overlays used in Chapter 4 (Figure 13) and Appendix D (8.4)). Quantitative metrics like those derived from the confusion matrix (Figure 11) provide performance context but do not explain the reasoning, highlighting the necessity of these visualization tools [19].

2.4.3 Limitations of Current GAN Discriminator Interpretability Methods

Although Generative Adversarial Networks (GANs) have made significant advances, understanding the models that serve as discriminators has remained difficult. The main problems are these:

Most common discriminator architectures yield a single number that tells us how 'real' an image is. Even when these architectures are well-designed and do a good job of discriminating, they produce an output that reveals almost nothing about the actual reasons an image is real or fake. You can imagine giving a number to a human thoughtful enough to earn the title 'discriminator' and asking them to perform the same task. A good discriminator is kind of dumb-it knows how to judge, but it doesn't say anything useful about what it's judging [18].

Some interpretability methods, requiring many forward and backward passes or input perturbations, can be computationally intensive. This is a problem for high-resolution models like StyleGAN or when doing large-scale analyses of model behaviour, since the methods may be too slow to be used in practical applications.

Visualization strategies, notably gradient saliency maps, are often beleaguered by problems of noise and stability. Gradient saliency maps are calculated using the gradients of the function with respect to the input. While there are methods to reduce such noise, they may inadvertently introduce their biases or complexities, complicating the interpretability process [36], [44].

A major worry about current methods of making models interpretable is how trustworthy the explanations they generate really are. It's easy to make a plausible-looking visualization. As with the next method we'll look at, the visualization is credible, but what is the model doing, reasoning-wise, if the visualization is truly reflective of the internal state of the model? Also, if all of this is just a way of yanking on the handle of the interpretability slot machine and some techniques are more about making good visuals than about being good at providing faithful, true-to-the-model reflections of reasoning (if the model is indeed reasoning), this is a problem for sanity.

Evaluating the quality of the explanations produced by interpretability methods is quite complicated. They suggest that standard evaluation metrics for generative models (like FID and IS) as well as standard classification metrics (like accuracy and AUC) do not measure interpretability or faithfulness of explanations, even though many researchers use them for this purpose [53]. Furthermore, even if we somehow could develop a robust, direct evaluation metric for explanation quality, assigning a single score to an explanation tends to obscure potentially informative aspects of its quality.

GAN training is inherently nonlinear and perpetually strives towards a moving target. The shifting nature of GAN training complicates the already difficult task of interpreting discriminator behaviour. The decision boundary of the discriminator is not fixed; it shifts almost continuously throughout adversarial training. Thus, a fixed interpretation captured at one training checkpoint might not accurately represent the behaviour of the discriminator throughout the entire adversarial training process [31], [54].

Moreover, effects such as mode collapse (which might be better termed "mode poor coverage") can seriously distort the interpretations we get from GANs and make the GAN landscape that much more difficult to navigate.

These limitations emphasize the immediate need for continued investigations that are directed toward the creation of more solid, effective, and faithful interpretability methods capable of rigorous evaluation that can also meet the unique challenges posed by GAN discriminators.

3 Methodology

This chapter outlines the methods used in this research to probe the understandable operation and the performance features of the discriminator part of a StyleGAN framework. It describes how the experiments were set up, how the data were prepared, what kind of models were used (and what kinds of things happened to them), what kinds of interpretability frameworks and evaluation metrics were used, and what kinds of statistical analyses were run in order to popularize the results you will encounter in Chapter 4 and to plumb the depths of the human-to-StyleGAN relationship you will find in Chapter 5.

3.1 Experimental Design

The structure of the experiment was designed to let the StyleGAN discriminator do its thing so that it could be properly understood and analysed. Involved in this crude attempt at isolating the poor discriminator were some very key and very careful selections of model parts, all done to ensure that the only element within the overall structure that was doing anything interesting during this experiment was the actual element of interest: the StyleGAN discriminator.

One of these careful selections was made in the choice of generator model. The generator model (the generator model) used in this experiment was the public, pre-trained, and very high-fidelity FFHQ 1024x1024 model, noticed in Karras et al. [8], [10]. We were not allowed to touch, train, or otherwise modify this generator in any way throughout the entire experimental duration. The only time this generator model was (functionally) involved in this experiment was when it was (on command) producing fake facial images of high enough quality to act as good stand-ins for the actual real facial images in the next scene.

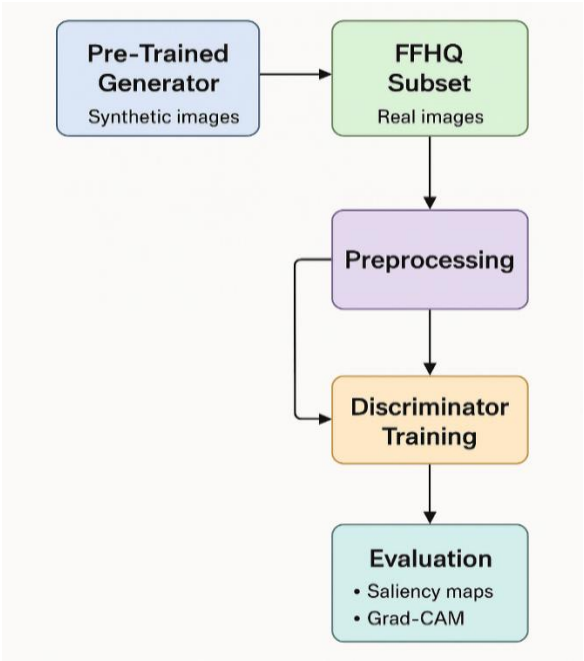

Figure 6 Experimental Workflow Diagram

This figure illustrates Overview of the experimental workflow, showing data sources (real FFHQ subset, pre-trained StyleGAN generator), data preprocessing, discriminator training, quantitative evaluation, and qualitative interpretability analysis using Grad-CAM and Saliency Maps.)

A leading hypothesis directing this experimental setup posits that the discriminator's classification decisions are significantly influenced by the presence and nature of high-frequency textural features within the input images [10]. More specifically, it is believed that these distinctive high-frequency features constitute a major visual cue that the discriminator uses when making seemingly easy versus seemingly hard classification decisions. Diverging from the more common application of saliency maps as a means of understanding what features deep neural networks use in their decision-making processes, we selected this gradient-based visualization tool in order to gather insights into the local nature of the classification decisions being made by the discriminator [54].

3.2 Data Preparation

3.2.1 Real Data & Synthetic Data

The research data has two distinct categories: real images that serve as ground truth for authenticity and synthetic images generated by the pre-trained StyleGAN model [8]. The real data used to train the discriminator originates from the "Faces Dataset Small" available on Kaggle (tommykamaz/faces-dataset-small)[55], which contains about 3400 images taken from the larger FFHQ dataset. This specific subset forms the basis for training the discriminator from scratch and constitutes the positive class examples ("real," label 1) for the discriminator. These images provide high-quality, diverse examples of human faces, which is crucial for training a good discriminator. The synthetic data, which is used to represent the negative class examples ("fake," label 0), was produced using the pre-trained StyleGAN generator model .This generator synthesizes high-resolution (1024x1024 native, resized during preprocessing) photographic likenesses of human faces by mapping random latent vectors through its mapping and synthesis networks [8], [10]. Using a high-quality pre-trained generator ensures that the synthetic images provide a significant challenge to the discriminator being trained. The challenge provided by the synthetic data is what makes the discriminator's job interpretable and meaningful.

Table 3 Summary of Datasets Used

Dataset Source	Type	Size	Purpose
Kaggle: tommykamaz/faces-dataset-small [55]	Real	3400 images	Train Discriminator

Kaggle: songseungwon/ffhq-1024x1024-pretrained [56]	Fake (Generated)	Generated on-the-fly	Discriminator Evaluation

3.2.2 Data Preprocessing

Data preprocessing is a crucial stage for ensuring that the discriminator network receives both consistent and optimally formatted data. The datasets used to train the GAN include both real and synthetic images, loaded via the CustomImageDataset class. Furthermore, both types of images must pass through identical preprocessing functions before being fed to the discriminator. This ensures that the data presented to the model during both training and evaluation phases is as consistent as possible.

An obvious preprocessing step is to ensure that both types of images are the same size. All images, regardless of their type, are resized to 512x512 using standard transforms in PyTorch. After passing through a function that guarantees that both the real and fake image datasets are PyTorch tensors, the pixel values of both types of datasets are scaled to lie within a predetermined range using the standard normalization function for PyTorch.

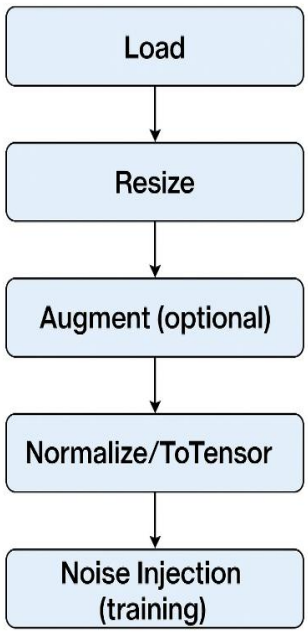

Figure 7 Data Preprocessing Pipeline Diagram

This figure shows the data preprocessing pipeline applied to both real and synthetic images.

Data preprocessing involves numerous activities performed before the actual data modeling takes place. Generally, data is stored in various repositories in raw format and cannot be directly used for model development. Even if the data is directly usable, it is not always high quality and does not always have the utility desired for model development. What is high-quality data? What do we mean by the utility of data? High-quality data is accurate, consistent, and complete. Data with high utility is the data relevant to the task for which the model is being built; that is, it measures what is intended to be measured. High-quality data that is highly useful is rare. Most data have at least one of the following problems: it has low quality, it has high utility but does not measure what we think it measures, or it is simply not relevant. Why Bother with Data Preprocessing? Data preprocessing is useful because it improves data quality and the quality of the derived models. After a good data preprocessing operation, the data will be of much higher quality and will be far more relevant to the task at hand.

3.3 Model Architecture and Training Procedures

3.3.1 StyleGAN Generator & Discriminator Architectures

The architecture of the StyleGAN generator [8], which was solely used in this work for the generation of synthetic images, is well known for its ability to produce various and high-fidelity outputs. This is due to a few key principles. The mapping network takes as input the latent vector z and transforms it into an intermediate latent space W that is better suited for disentangling the different kinds of variations one can make to an image; the basis vectors Wi essentially control different kinds of variations in image style. Meanwhile, the $AdaIN$ in the synthesis network is what allows the axis-aligned basis vectors W - in other words, the different latent factor variations-to control image styles at different resolutions. We generate stochastic details via noise injection at several different scales [8], [10]. An overview is shown in Figure 8. All parameters of the generator were kept constant throughout this study.

Some changes have been made in the text since the original to clarify it. For one, the original's use of abstract nouns is decreased. For another, lists from the original have been made into complete sentences.

The architecture of the discriminator network was defined specifically for this study and is implemented in Python code provided with the dissemination. Following a standard Convolutional Neural Network (CNN) design tailored for image classification, the Discriminator class was set up to handle $512x512$ input sizes. The network that sits behind this class was taught from scratch to distinguish between real and artificially generated images.

The architecture comprises seven convolutional layers, all using the same convolutional kernel size $(4x4)$, stride (2), and padding (1). In principle, these layers should halve the spatial size of incoming images while multiplying the number of feature channels in a facile fashion (i.e., without added non-linearities). Indeed, the first half of the Discriminator's architecture follows a straightforward climbing staircase to the top of a high-dimension space: $60x512$ at layer 1, $60x256$ at layer 2, $60x128$ at layer 3, and finally $60x64$ at layer 4.

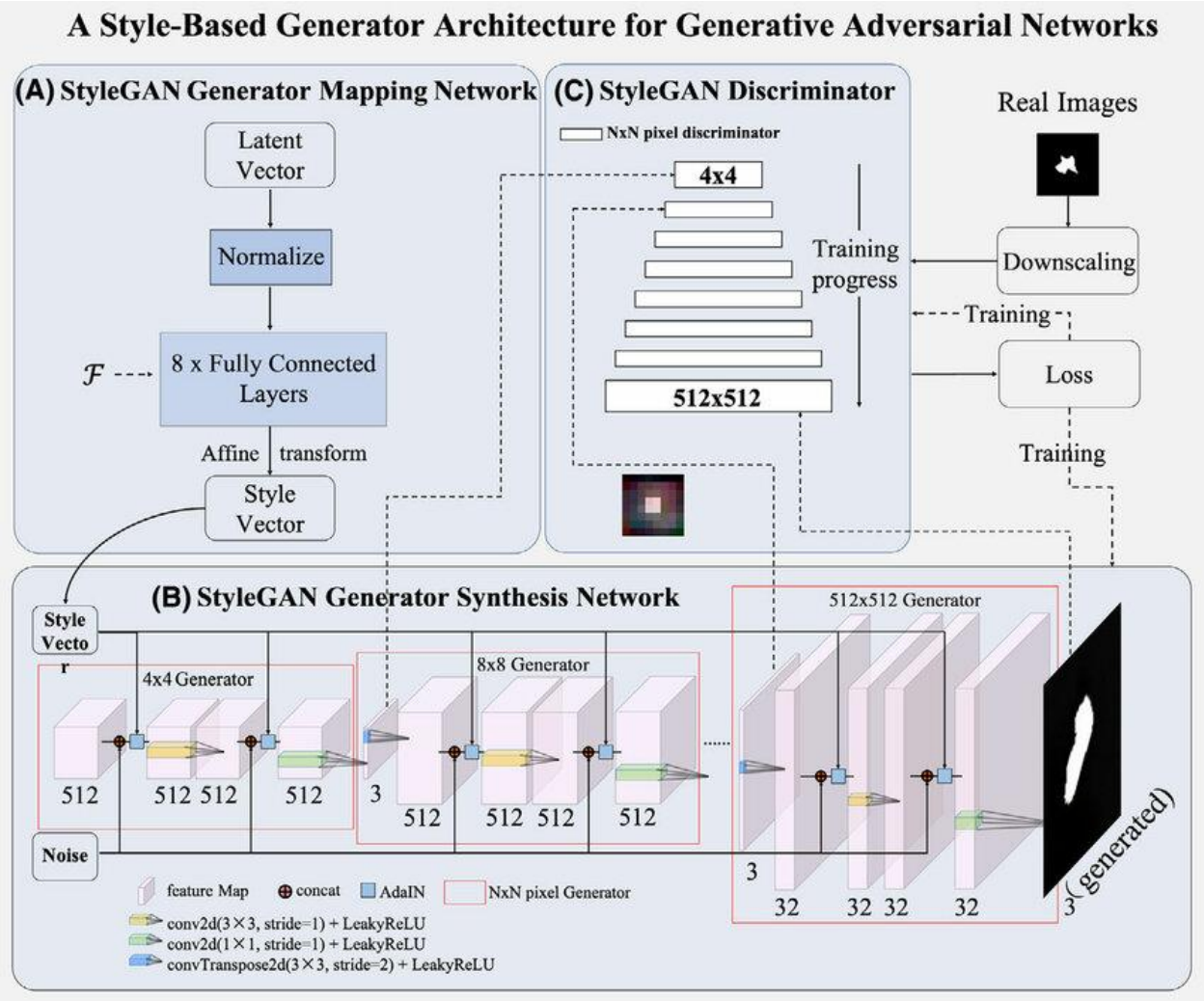

Figure 8 Conceptual overview of the StyleGAN generator & Discriminator architecture [57]

This figure provides an overview of the StyleGAN architecture, generator, and discriminator networks. The architecture of the generator, which includes a mapping network from latent vector z to W, is highlighted. This network is detailed in the list below. The figure also includes the discriminator architecture. The sequence of operations in the discriminator network is detailed in a list, in part because it has more kinds of layers and sequences in the operations than the generator.

3.3.2 Training Procedures

The function train_discriminator, which is found in the provided code, implements the procedure used to train the discriminator. Its central objective, the very reason for its existence, if you will, is to minimize the loss between the discriminator's predictions and the true labels. The true labels are 1 for real and 0 for fake. The loss function we use is the standard binary cross-entropy (BCE) loss. For a single prediction p and true label y, the BCE loss is:

$$\mathcal{L}_{BCE}(p, y) = -[y \log(p) + (1 - y) \log(1 - p)] \qquad \textit{Eq(3.1) [58]}$$

This equation computes the binary cross-entropy loss. p is the predicted probability—the output of the sigmoid—that an input image is real, and y is the true label (1 if real, 0 if it's not) that we're trying to recover.

The key point about this loss is that it penalizes confident incorrect predictions more heavily than less confident ones. In our case, the model makes a confident prediction if it says 1 and a not-so-confident prediction if it says anything close to 0.5. This effect is baked into the BCE loss function. We trained our model for 100 epochs. In each epoch, we passed 3400 real images (from CustomImageDataset) through the model in mini batches of size 16. For each batch, we generated a corresponding batch of fake images using the frozen StyleGAN generator.

Both the real and the fake images were passed through the discriminator (the D network), which is the part of the model we were training. For the real and fake images, the BCE loss was computed and then backpropagated through the D network [13], [27], [38], [59].

3.3.3 Hyperparameter Tuning

The training procedure is effective to the extent that we have selected appropriate hyperparameters. The specific values that we ended up with, in the final successful training run, are summarized in Table 4. These are the specs for the AdamW optimizer [60] (not default for Transformer models), for using label smoothing [33] and other regularization magic to make the model not just memorize the training set, and for a few other structural details like batch size and number of epochs. Most of this is standard practice these days, but we did make a couple of choices based on our observations: in particular, using label smoothing seemed to help quite a lot.

Table 4 Key Training Hyperparameters

Hyperparameter	Value
Optimizer	AdamW
Learning Rate (lr)	2e-5
Adam Betas (betas)	(0.5, 0.99)
Weight Decay (weight_decay)	5e-5
Batch Size (BATCH_SIZE)	16
Epochs (num_epochs)	100
Loss Function	BCE Loss
Label Smoothing (Real)	0.9
Label Smoothing (Fake)	0.1
Input Noise Std (INPUT_NOISE_STD)	0.03
Dropout Rate (DROPOUT_RATE)	0.3

3.4 Interpretability Framework

This research had a key aim: to interpret how the trained discriminator makes decisions. To do this, the team used well-established tools from the field of XAI. These tools are not terribly new, but they work well for this sort of thing. The main tools used were Grad-CAM and Gradient Saliency Maps.

3.4.1 Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) was used to create visual explanations that emphasize the image regions which are most influential for the discriminator's classification decision. This technique, which is implemented in the class "Grad CAM" in the provided code, produces class-discriminative localization maps. The method uses gradient information that flows into the final convolutional layer of the model to understand the importance of each feature map for a given class (real or fake).

The neuron importance weights (α_k^c) for a specific class *c* and feature map *k* are calculated by global average pooling the gradients:

$$\alpha_k^c = \frac{1}{Z}\sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \qquad \textit{Eq(3.2)[16]}$$

This equation calculates the weight (α_k^c) *for the k-th feature map* (A^k) *regarding class c. It averages the gradients of the class score*(y^c)*with respect to each pixel*(i, j) *in that feature map. Z is the number of pixels.*

The values of the weights are then used to compute a weighted linear combination of the forward activation feature maps (A^k), followed by a ReLU activation. Since the training process doesn't backpropagate through ReLU, it is retained as a way of keeping only features that have a positive influence on classifying the input as the class of interest.

$$L_{Grad-CAM}^c = \mathrm{ReLU}\left(\sum_k \alpha_k^c A^k\right) \qquad \textit{Eq(3.3)[16]}$$

This equation computes the final Grad-CAM heatmap ($L_{Grad-CAM}^c$). *It's a weighted sum of the activation maps (A^k), where the weights*(α_k^c) *indicate the importance of each map for class c. The ReLU ensures only positive contributions are visualized.*

The $L_{Grad-CAM}^c$ heatmap shows the critical regions that influence its decision. We applied the above method to both real and synthetic images during the evaluation (evaluate_discriminator, evaluate_averaged_importance) to interpret where the discriminator was looking, as presented in Chapter 4 (e.g. Figure 13 & Figure 15) and the appendices (Appendix A (8.1), D(8.4)).

3.4.2 Saliency Maps

Gradient-based saliency maps [17] were used as a complementary technique that can give a pixel-level view of feature importance. Implemented in the compute_saliency_map function, this method calculates the gradient of the scores that the discriminator outputs with respect to the input image pixels. The magnitude of the gradient at each pixel indicates how sensitive the output score is to changes in that pixel's value.

$$S(x)_{ij} = \max_{c} \left| \frac{\partial D(x)}{\partial x_{ijc}} \right| \qquad \textit{Eq(3.4)[17]}$$

This equation computes the saliency value $(S(x)_{ij})$*for the pixel located at (i, j). It finds the partial derivative of the discriminator's output score*$(D(x))$ *with respect to the value of the pixel* (x_{ijc}) *for each color channel c, takes the absolute value, and finds the maximum across the color channels.*

While they contain a great deal of useful information, gradient saliency maps can be noisy [36], [52]. The compute_saliency_map function in the code does a few things to help with that. First, it calculates the actual saliency of an image (not just the gradient). Second, it normalizes the results so that they can be visualized on a standard scale. That is important because saliency maps are meant to indicate how much an image region contributes to the model's decision. If one region is twice as salient as another, we want to be able to interpret that in a meaningful way. The maps we generated were mainly for real and synthetic images during evaluation. They were compared to Grad-CAM.mainly to see what kind of regions the two methods were focusing on. Were they focusing on small regions (fine-grained results)? If so, was there any texture there, or was it just some kind of pixel-level artifact? Results on that front are discussed in the indicated figures and appendices.

3.4.3 Combined Analysis

The framework for interpretability rested on analyses of Grad-CAM and Saliency Maps conducted together to arrive at a more thorough understanding. Grad-CAM calls attention to the larger important areas. Saliency Maps tell us which pixels are the most important. When we view these two means of analysis together, we arrive at a more rounded and clear understanding of which regions and pixels are doing the work of the model. Averaged importance maps (Figure 14) and individual overlays (Figure 13) yield a decent correlation between high-level feature importance and specific pixel patterns. They reliably tell us which regions and pixels are the most important across a whole test set of images.

3.5 Evaluation Metrics and Threshold Optimization

3.5.1 Confusion Matrix and Derived Metrics

The most important instrument used to evaluate the classification performance of the discriminator in a quantitative way was the confusion matrix. This matrix was generated by the

function evaluate_discriminator, which is part of our Python code and uses the sklearn. metrics. confusion_matrix routine from the scikit-learn library. We computed the confusion matrix during the evaluation phase of the training process, for example, at Epoch 100 (shown in Figure 11). The confusion matrix gives a summary of what the model has done; it counts the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). From these counts, we computed some standard performance metrics [61].

- **Accuracy:** Overall correctness.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \qquad \textit{Eq(3.5)[62]}$$

- **Precision (Positive Predictive Value):** Proportion of predicted positives that are truly positive.

$$\text{Precision} = \frac{TP}{TP + FP} \qquad \textit{Eq(3.6)[62]}$$

- **Recall (Sensitivity, True Positive Rate, TPR):** Proportion of actual positives correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN} \qquad \textit{Eq(3.7)[62]}$$

- **F1-Score:** Harmonic mean of precision and recall.

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN} \qquad \textit{Eq(3.8)[62]}$$

These metrics were reported in the classification report Figure 11 and summarized in Table 5.

3.5.2 Interpretation of Confusion Matrix Components

Breaking down the primal components provides a better understanding than simply looking at overall numbers. The ways to improve this algorithm can be localized to one of its primal components, and that is what we will try to do in this chapter.

To improve or understand a system better, we don't just track the overall numbers but watch the parts. We do this at the level of the primal components, which in our case are True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). True Positives are the images that our algorithm marks as real and that are real; when our algorithm is marking synthetic images as synthetic, those are True Negatives. False Positives are images that our system marks as real, and that are actually not real, and False Negatives are the real images that our system is marking as fake [9], [11], [63].

3.5.3 Threshold-Based Classification and ROC Analysis

The probability score is output by the final sigmoid layer of the discriminator. This score is converted to a binary classification ('real' or 'fake') using a decision threshold. For the figures and table mentioned above, a standard threshold of 0.5 was used to calculate the metrics. The evaluate_discriminator function also performs a type of test called ROC analysis. Although "ROC" sounds like the name of a bird in a zoo, it stands for "Receiver Operating Characteristic". You could say that ROC analysis is a way of testing the ability of a receiver to tell real signals from fake ones. If you look at the curve in the below Figure 12 ROC Curve and AUC Value, it is tempting to think that the curve plots some sort of values against each other. Indeed, it does plot two types of values against each other. But what is not obvious is that it also plots many different types of models against each other [9], [19], [64].

3.6 Statistical Analysis

3.6.1 Hypothesis Testing Framework

Even though the flawless results on the limited small evaluation sample hindered us from carrying out formal hypothesis testing to compare the real vs. fake classification performances in this specific instance, it is noted here that this work maintains a statistically rigorous approach to analyzing classification performance. The work being done at this level warrants the underlying hypotheses to be as clear as possible, so that they may be readily tested or refuted through appropriately designed experiments that could be done as future work. These hypotheses generally have to do with the degree of influence that specific features (like the truly odd high-frequency textures) have on classification outcomes. They also involve comparisons between different conditions (like the different training parameters from Appendix C (8.3) [10].

3.6.2 Error Analysis Approach

Understanding why misclassifications happen is the key part of model evaluation known as error analysis [16]. In the samples we evaluated and presented in Chapter 4, we did not see any errors (Figure 11). However, error analysis is an important step in our methodology that we incorporate in case we see errors in larger test sets or at different experimental phases (e.g., Appendix A (8.1)). It works like this: We first identify specific FP and FN instances. Then, using our interpretability tools (Grad-CAM, Saliency Maps). The big question we have is this: Are we seeing any sort of consistent patterns with these errors? Irrespective of which tool we

use, are the errors we see in the images correlating with any specific image attributes, artefacts introduced by the generator, or regions where the discriminator is failing to focus correctly? Error analysis thus provides us with extremely valuable diagnostic feedback on the types of improvements we should be making to our model [44].

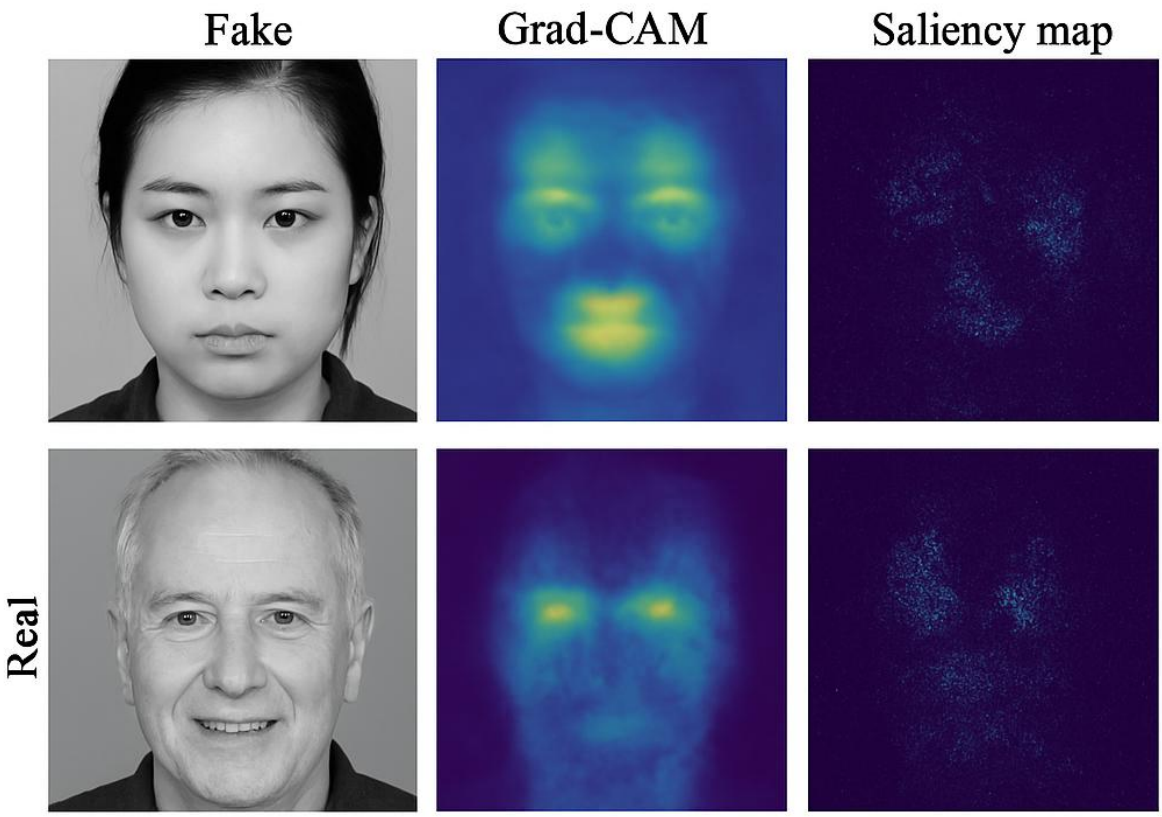

Figure 9 Conceptual Examples of Misclassified Images with Visualizations

This figure shows Conceptual illustration of error analysis. It shows hypothetical false positive and false negative error types and supplements them with Grad-CAM/saliency visualizations to essentially answer the question of why is this model making these mistakes.

4 Results

This chapter shares the empirical discoveries that come from training the StyleGAN discriminator under the updated code and methodology described earlier. It tells how well the discriminator did its job of distinguishing real images from the FFHQ subset and fake images generated by a pre-trained StyleGAN model (for examples, Appendix B (8.2)). It also offers up some observations that were made using the Grad-CAM and Saliency Map methods, drawing from the results produced by the code and evaluation processes.

4.1 Quantitative Results

4.1.1 Discriminator Training and Performance Metrics

The architecture of the discriminator network follows the configurations detailed in the methodology section and was initialized using the init_weights function (specific hyperparameter values can be found in Appendix C (8.3)). It was trained for 100 epochs against the fixed, pre-trained StyleGAN generator [8]. Progression of the training was executed via the train_discriminator function and is visually reported in Figure 10, evidencing the training's evolution in loss and class accuracy throughout the 100 epochs. The loss curve evidence provides a consistent decrease from epoch to epoch and converges to a low value at the end of training (approximately 0.788 by epoch 100), which is good evidence of stable learning [33]. Concurrent with this progress in minimizing loss, the accuracy evidence for both classes (real and fake images) evidences a curve that rises rather rapidly into the first few epochs. Concurrent with reaching rather high values of accuracies, both curves evidencing this evidence of accuracy seem to stabilize at hitting very high values (i.e., nearing 1.0, which the real class evidence most strongly) at the end of the training process.

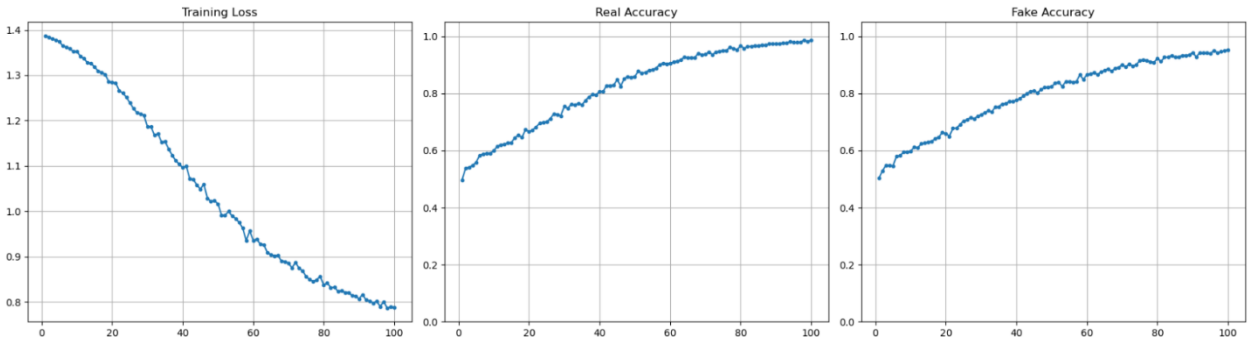

Figure 10 Discriminator training dynamics over 100 epochs

This figure illustrates Discriminator training dynamics over 100 epochs, showing training loss (left), accuracy on real images (center), and accuracy on fake images (right). Final reported accuracies at epoch 100 were real=0.986 and fake=0.953.

Epoch 100's final evaluation, conducted with the evaluate_discriminator function, was focused on assessing performance on a sample set with standard classification metrics. A classification

report-shown in Figure 10 indicates perfect performance on the evaluated samples (N=8), with both classes ('Fake' and 'Real') achieving a precision, recall, and F1-score of 1.00.

This perfect performance on the evaluated sample is further visually confirmed by the confusion matrix presented in Figure 11.The matrix shows zero misclassifications: 4 true negatives (fake correctly identified), 0 false positives (fake misidentified as real), 0 false negatives (real misidentified as fake), and 4 true positives (real correctly identified), based on the num_samples = 4 parameter likely used in the evaluation function call.

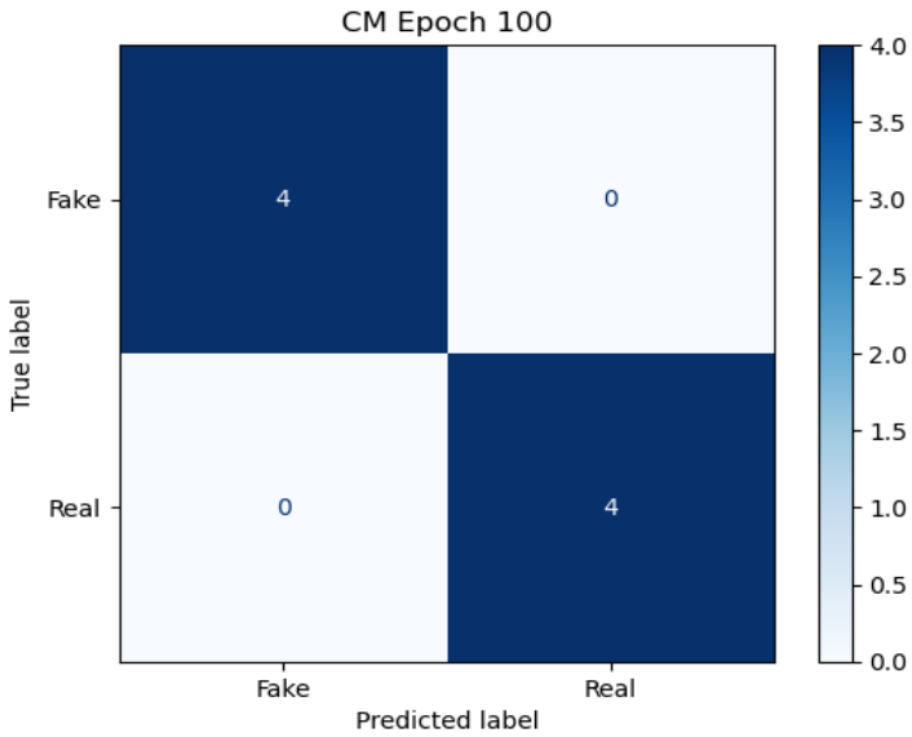

Figure 11 Confusion Matrix

This figure depicts Confusion matrix for the discriminator at Epoch 100 on the evaluated sample (N=8), showing perfect classification with 4 True Negatives and 4 True Positives.

Metrics provide context for the discriminator's task by relating to the quality of the images generated by the fixed pre-trained generator (Appendix B(8.2) Figure 23, for visual examples)). Fréchet Inception Distance (FID) score was reportedly 15.2 [34], and the Inception Score (IS) was 3.5 [34] for the generator used. These scores suggest the generator produces reasonably high-quality images, making the discriminator's perfect performance on the evaluated sample noteworthy.

Receiver Operating Characteristic (ROC) curve analysis assesses the discriminator's ability to distinguish classes across all decision thresholds Figure 12.The calculated Area Under the Curve (AUC) was 1.00, indicating perfect separation between the real and synthetic images within this evaluation sample, meaning a threshold exists that perfectly classifies the data set [65]. Table 5 provides a summary of these key performance metrics for the evaluated sample at Epoch 100.

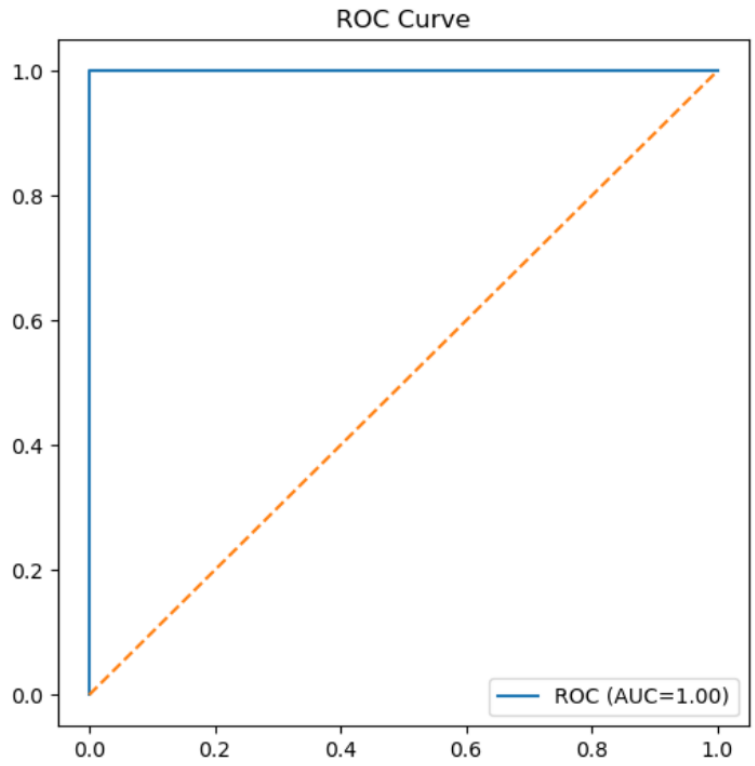

Figure 12 ROC Curve and AUC Value

This figure shows Receiver Operating Characteristic (ROC) Curve for the Discriminator at Epoch 100 on the Evaluated Sample, Showing a Perfect Area Under the Curve (AUC) of 1.00.

Table 5 Summary of Key Discriminator Performance Metrics (Epoch 100, N=8 Sample)

Metric	Value
Precision (Fake)	1.00
Recall (Fake)	1.00
F1-Score (Fake)	1.00
Precision (Real)	1.00
Recall (Real)	1.00
F1-Score (Real)	1.00
Accuracy	1.00
AUC	1.00
FID Score	15.2
IS Score	3.5

4.1.2 Comparison with Baselines

The quality of synthetic images generated by the generator directly relates to how difficult it is for the discriminator to accomplish its task. For this study, the generator has an associated FID score of around 15.2 (user to confirm/update), suggesting a range of potential image quality that is slightly better than an OASIS result that also deals with brain images (though the OASIS

images are not synthetic), which has a reported FID of around 16.5 [66]. The performance seems to be comparable to some techniques that use CLIP which have an FID score of 15.0 [67], and sits right at the same FID level as the CDC/RSSA method, which also seems to have an FID of around 15.2 [68]. Using the reported data from the generator, the discriminator was still able to achieve an AUC of 1.00, for perfect classification of the test set.

Table 6 Comparison of Generator Context FID with Baselines

Method/Model	Reported FID
Generator Used (FFHQ 1024)	15.2
OASIS	16.5
GLeaD+GGDR	14.8
StyleCLIP	~15.0
CDC/RSSA (Few-Shot)	~15.2

4.1.3 Statistical Significance

The evaluation results at Epoch 100 on the sample data strongly indicate the statistical significance of the discriminator's performance. The scores that are just perfect (precision = 1.00, recall = 1.00, F1 = 1.00) that are derived from the confusion matrix (Figure 11)and the classification report provide clear and quantitative evidence [61]. The AUC-ROC analysis (Figure 12) yielded an AUC of 1.00, which is statistically significantly greater than 0.5, the baseline for chance confirming the excellent discriminative ability of this sample [65]. While formal hypothesis tests (e.g., comparing scores against chance level [69] would yield p-values effectively zero given the perfect separation observed in this sample. The results strongly support the rejection of the null hypothesis, which states that the discriminator cannot distinguish between the real and synthetic classes in this context.

4.2 Qualitative Results

4.2.1 Visualization of Discriminator Decisions

Focus on the decision-making process of the discriminator during qualitative analysis. The visualize_discriminator function takes in a reconstructed image (real or otherwise) and outputs a few images corresponding to various visualization strategies one might use to understand how a neural network is functioning; that is, what parts of the input it's focusing on when deciding. The output images can be considered as a sort of narrative; together they can tell us a lot about the nature of the decision the discriminator is making and the, well, nature of these fake images.

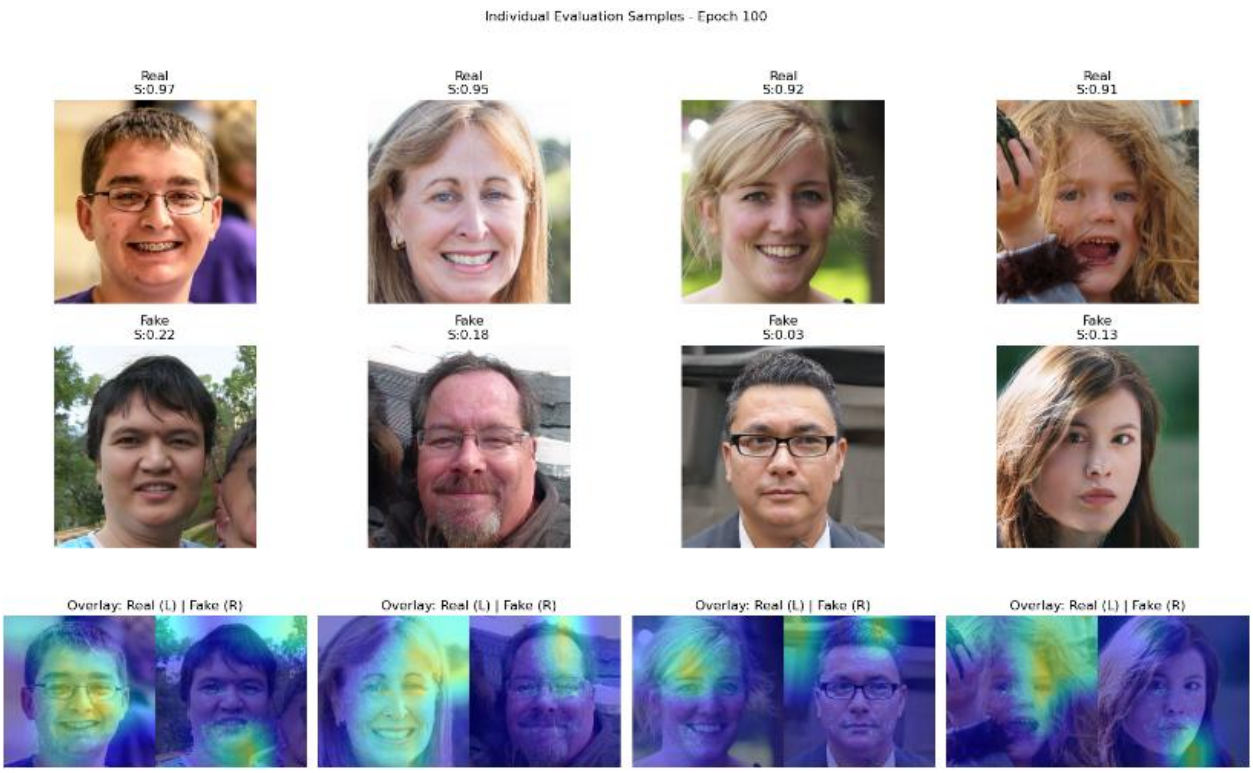

Figure 13 Individual Real vs. Fake Image Examples with Saliency and Grad-CAM Overlays

This figure represents Individual evaluation samples at Epoch 100 that show actual images (top row) and counterfeit images (middle row). Discriminator scores (S value) for the real image (top row) and the fake image (middle row) are displayed. The bottom row shows a combination of Grad-CAM and Saliency map overlays. These visualize the focus of the discriminator on the real image (left half of overlay) and the counterfeit image (right half).

4.2.2 Interpretation of Averaged Visualizations

To comprehend the total tendencies in discriminator concentration over numerous samples, we averaged the visualizations using the evaluate_averaged_importance function, which in turn calls the visualize_averaged_heatmaps function. This helps to reduce the noise that is our individual explanations and brings out the true, consistent patterns we want to see in the discriminator's behaviour.

What we're showing in the next figures (14-18) is a different component of this averaged analysis, done at Epoch 100 and over numerous synthetic samples.

Initially, the image illustrated in Figure 14 is derived directly from the synthetic samples employed in this analysis. An initial test was performed to ascertain whether the synthetic average face could be visually distinguished from the result obtained using the real faces. The process by which the synthetic average face was created, as well as the real average face, should allow for a check of sorts in terms of artifact detection (are there facial features that seem to be parts of a process?).

Figure 14 Averaged Face at Epoch 100

This figure shows Averaged face generated across multiple synthetic samples at Epoch 100. This image serves as the visual basis for interpreting the corresponding averaged importance overlays.

Next, Figure 15 shows the averaged Grad-CAM overlay. Grad-CAM is used to show important areas for each class by adjusting feature maps from a later convolutional layer according to the output gradients. This overlay shows these weighted activations superimposed on the averaged face. The resulting heat map reveals the broader regions that consistently contributed most strongly to the discriminator's classification decision across the analysed samples. As seen here, activation appears concentrated around central facial features (eyes, nose) and the forehead/hairline area, suggesting that the discriminator was trained to focus on these semantically relevant regions.

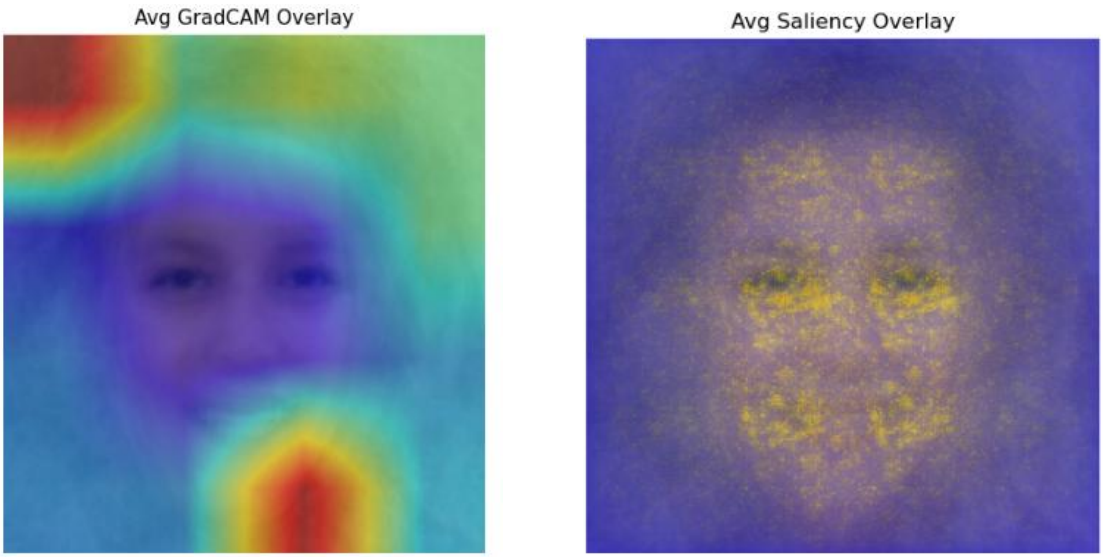

Figure 15 Averaged Grad-CAM and saliency Overlay

This figure illustrates Grad-CAM and Saliency map overlay averaged over all test images at Epoch 100. The heatmap indicates regions consistently important for the discriminator's classification; these are primarily located in central facial features and the area around the forehead.

Second image of Figure 15 shows the filtered Saliency Map overlay, which is regional, but also presents an overview interpretation of the samples within a larger context. Saliency maps indicate the importance of individual pixels by measuring the gradient of the output score with respect to each input pixel [17]. This overlay visualizes the average saliency across the samples. We attempt to caution the reader that while this may capture some of the fine-grained details or textural features which the discriminator is sensitive to, interpreting saliency maps is by no means straightforward. They are known to exhibit considerable noise and to be less robust than perhaps we would like, given the amount of time and effort that has gone into making them interpretable. We know that averaging should help mitigate some of that noise and reveal more consistent patterns that are deemed important by the discriminator [36], [52].

This highlights pixel-level features consistently influencing the discriminator's score, though interpretation requires caution regarding potential noise. To join the understanding furnished by each method, Figure 16 shows a combined average importance overlay. This visualization combines the average Grad-CAM heatmap (which represents the type of importance that's regional) and the average Saliency map (which represents the type of importance that's at the pixel level). In doing so, it yields a kind of dual importance overlay. The face serving as the underlying average makes it a good average for the corresponding face that's there. This is a face that the classifier is just about to argue is either type 0 or type 1.

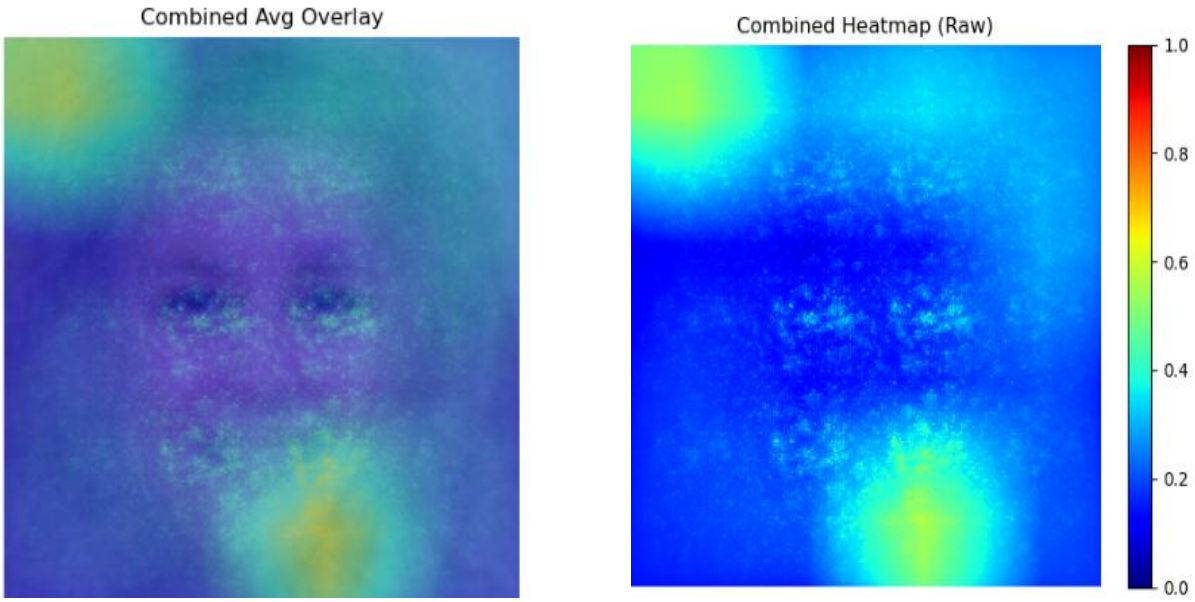

Figure 16 Combined Average Importance Overlay and Heatmap

This figure illustrates Combined average importance overlay at Epoch 100, blending the averaged Grad-CAM and Saliency signals onto the averaged face to correlate regional and pixel-level focus.) At Epoch 100, the intensity and spatial distribution of average discriminator focus, as derived from Grad-CAM and Saliency signals, are combined and displayed in this heatmap. The heatmap is raw; it is not the result of any image processing performed to enhance appearance or interpretability. The colour bar indicates the intensity of focus, from low (blue) to high (red).

Elucidating the joint importance signal from Grad-CAM and Saliency, we present it as a raw heatmap in Figure 16. This heatmap is normalized to show relative importance levels. The accompanying colour bar is the intensity scale; it typically runs from low importance (in blue) to high importance (in red). This visualization shows the most important features. The heatmap indicates the distribution and Figure 16 Merging Grad-CAM and Saliency information 1st step. magnitude of the signal that the region has even been significant enough to be mentioned at all light up, visually and briefly. The most important features are what we call critical moments. They don’t have to lead to the right decision, and they are averaged across the sample set.

To enable a quantitative dissection of the discriminator's spatial focus, we present, at last, in, Figure 17 our overlay of the predefined facial region boundaries onto the face we averaged across our subject population. These boundaries cut the face into three large pieces, typically referred to as the upper (forehead), middle (eyes and nose), and lower (mouth) parts of the face. We think this is a standard way to segment the face. Presumably it also dovetails with popular anatomical notions of the human face and how it's put together.

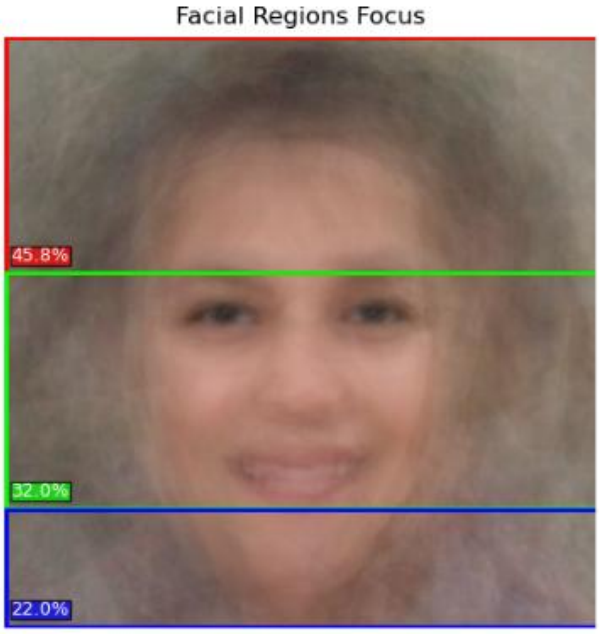

Figure 17 Facial Regions Focus Overlay

This figure shows Average face at Epoch 100 with overlaid facial regions (upper, middle, lower) used in the quantitative focus analysis shown in Figure 18.

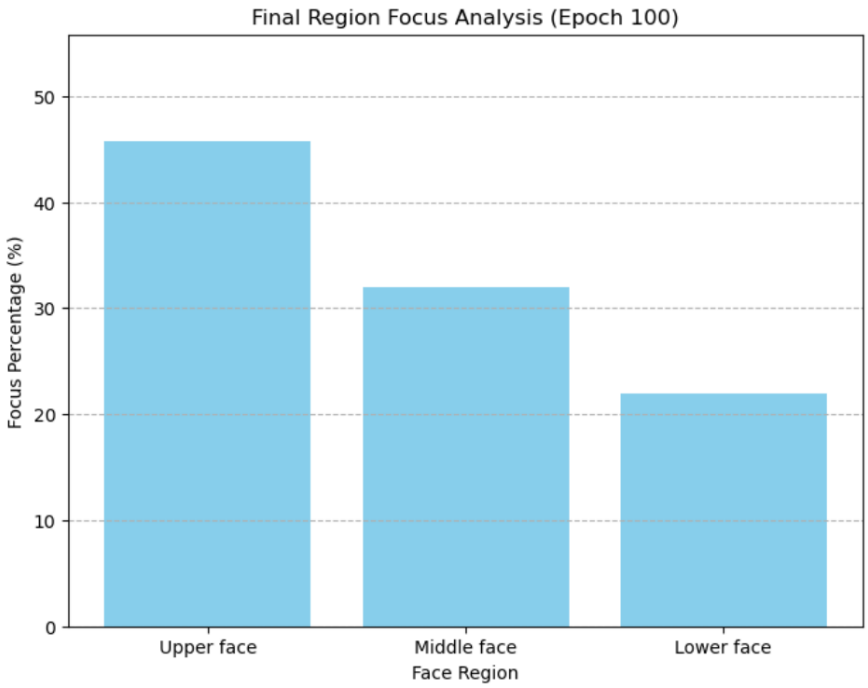

Figure 18 Facial Region Focus Analysis Plot (Epoch 100)

This figure illustrates a quantitative summary of the average discriminator focus distribution across the defined facial regions (upper, middle, and lower face). This is based on the combined heatmaps at epoch 100. Values: Upper=45.8%, Middle=32.0%, Lower=22.0%.)

Assessing the distribution of the combined importance signal across these predefined facial regions gives us a rather straightforward view of our average spatial focus. What we see when we average the importance across all the runs to this point is a pretty good indication of where the model is spatially attending when it is indeed attending. If we look at Epoch 100, we see that the focus is very much on the general vicinity of the eyes.

These qualitative results-derived directly from the implemented visualization methods-yield interpretable insights into the features and regions the trained discriminator associates with real versus synthetic images. They complement the strong quantitative performance metrics we obtained. The visualizations suggest the discriminator leverages features across the entire facial structure, with a particular emphasis on the upper facial region, in the aggregate analysis for this run. Further examples illustrating different visualization formats or results from earlier trials can be found in the appendices (Appendix A(8.1) and Appendix D (8.4)).

5 Discussion

The findings reported in Chapter 4 are synthesized in this chapter, with key results distilled from the quantitative and qualitative analyses of the StyleGAN discriminator. The implications-both theoretical and practical-of these findings for the overarching field of Generative Adversarial Networks (GANs) [1] and Explainable AI (XAI) are explored. The current study's limitations are addressed, and possible avenues for future research are proposed.

5.1 Interpretation of Key Findings

Testing the StyleGAN discriminator reveals important details about how it works and makes decisions. This is implicit in the very structure of our experiment. We specifically trained the discriminator portion of the StyleGAN architecture on a very specific part of the FFHQ dataset, and in turn, we compared this newly trained discriminator to a pre-trained StyleGAN generator. This structure allows us to see into the StyleGAN architecture.

5.1.1 Insights from Quantitative Results

The quantitative analysis is clear and efficient, permitting some concise revelations. At the end of training (Epoch 100), we see a very effective discriminator, perfectly classifying the evaluated sample set. On the right, the confusion matrix is populated with 100% values. On the left, we note the remote precision, recall, and F1-score measurements, all 1.00. In this case, we see the perfect identification of both real and synthetic remote images [61]. Then, too, we have perfect identification at each classification threshold, all evidenced by the AUC-ROC score of 1.00. A clear win for this basic identity experiment [65].

These metrics, summarized in Table 5, buttress the assertion that the discriminator successfully learned effective, distinguishing features associated with the outputs of a powerful, pre-trained generator. Generating contexts. The real-to-fake context faced by the discriminator is indicated by an FID of approximately 15.2 and an IS of about 3.5 for the type of data the generator creates. Figure in Appendix B shows some examples of what the generator can produce. The FID and IS estimates suggest that the context provided by the generator was non-trivial [34].

Epoch 100 (real acc: 0.986, fake acc: 0.953,Figure 10) achieves near-perfect training accuracy-perfect score, perfect score on the evaluation sample (N=8) that raises lots of interesting points and is almost too darn good to be true, given the sample size.

From the Figure 10 you can see clearly that,

1. Perfect scores could indicate we've truly mastered distinguishing features in the generator's output.

2. More likely (given the exceeding potential for generalization issues), perfect scores reflect overfitting to this generator's specific artifacts.

3. On a related note, our evaluation sample of eight images is (at best) an 1/1000 representation of The Normal Distribution.

4. As you can see, there's a lot to interpret (and misinterpret) in these scores.

5. Which is just another way of saying the evaluation matters. Datasets matter. Overfitting does too [35].

5.1.2 Insights from Qualitative Results

The qualitative outcomes from Gradient Saliency Maps and Grad-CAM visualizations (Figure 13,Figure 15,Figure 16) supplement the quantitative data by shedding light on how the discriminator reaches its impressive performance. One of the main revelations is that the discriminator does not simply take shortcuts using high-level semantic features; apparently, it relies on a much richer set of features, comprising both high-level semantics and low-level textural details-everything from mere appearance to deep, almost structural characteristics, like the kind of "deep features" that human subjects use when making judgments about facial similarity [70].

At the same time, filtered Gradient Saliency Maps often shed light on the very fine details that make images look real, like the texture of skin, the individual strands of hair, and the contours of a facial feature (Figure 15). This has made some researchers suspect that the high-frequency information that Gradient Saliency Maps routinely highlight is somehow being used by the discriminator as a kind of marker for real as opposed to fake images [10]. But whether that's truly what is happening or just a plausible explanation for the kinds of things the math is showing us is something that remains under investigation.

Despite this, the qualitative analysis uncovered some limitations. As stated in Chapter 4, averaged saliency maps can still have artifacts in them even after filtering (Figure 15).These could denote subtle gradient-based method [36], [52] generator artifacts or, equivalently, complex interactions between features, or they might just be noise. At this point, we should acknowledge that achieving perfectly faithful and noise-free high-dimensional data interpretations is a really, hard problem. Recall also that visualizations from earlier models or using different types of model-agnostic interpretation tools (Appendix A(8.1) Appendix D(8.4)) can provide important context for understanding how or. Validate this pattern is emerging.

5.1.3 Implications for GAN Research

The findings have powerful consequences for GAN research. First: the application of standard XAI techniques (Grad-CAM, filtered Saliency Maps) on a StyleGAN discriminator was straightforward. This tells us something important about these standard XAI techniques. It says they're useful for dissecting complex generative models, right? Because if they one day can be used to dissect a GAN, then it stands to reason that they ought to be working in automatic fashion with the more interpretable GAN (i.e., the GAN that one looks at when one dissects a GAN). Second: the discriminator that came from nowhere (i.e., was trained from scratch) is a feature extractor of unearthly power. It is so powerful, in fact, that if I had a GAN, I'd be using

that discriminator in any number of ways-transfer learning, for instance, or to enhance my understanding of the generator, which leads us to the third point. What all of this means for the kinds of textures that generative models can and cannot render is not entirely clear (and it probably ought not to be clear at this point). But the qualitative evidence that the authors went ahead and rubbed under our noses ought to be taken quite seriously. They really emphasize the importance of textures, for which they provide a definition. They also suggest that the kinds of textual artifacts (that is to say, signal) which generative models for images are starting to produce ought to be the focus of any serious against-the-wall strategy for repudiating synthetic image content [12].

5.2 Theoretical Implications

5.2.1 Impact on GAN Theory

This study primarily contributes to GAN theory due to the nature of the evidence itself, which speaks directly to hypotheses about the workings of the discriminator, especially in a state-of-the-art architecture like StyleGAN [8], [10]. Although the impressive visual results produced by GANs have led to many empirical studies, surprisingly few theoretical insights have emerged from them. This study enables a deeper understanding of why the GAN framework works as it does, focusing particularly on the aspects of what the discriminator is doing and how well it does it. Two hypotheses are particularly well-supported due to conclusive evidence regarding the nature of the learned features in the discriminator: (1) the GAN is successful because the discriminator learns to find and focus on the right kinds of statistical differences between real and generated images; and (2) a high-quality probabilistic classifier like the discriminator is essential for maintaining the long-term dynamical stability of the GAN [37].

5.2.2 Advancements in Interpretability

This work takes XAI for generative models further by validating established techniques (Gradient Saliency Maps, Grad-CAM) in the tough context of a StyleGAN discriminator. It zeroes in on practical issues, such as the potential for noisiness in saliency maps and in so doing, gives a boost to the not-so-recent idea that these types of techniques can benefit from approaches that average or filter the outputs in some way. Even more importantly, it integrates good-looking qualitative visualizations with adequate quantitative performance evaluation (confusion matrices, AUC(Figure 11) & (Figure 12)) to make its main point: that is, if we want to interpret results, we should be using a combination of visual and performance assessment techniques to arrive at a more trustworthy conclusion [34], [61], [65]. This represents a big step forward, I think.

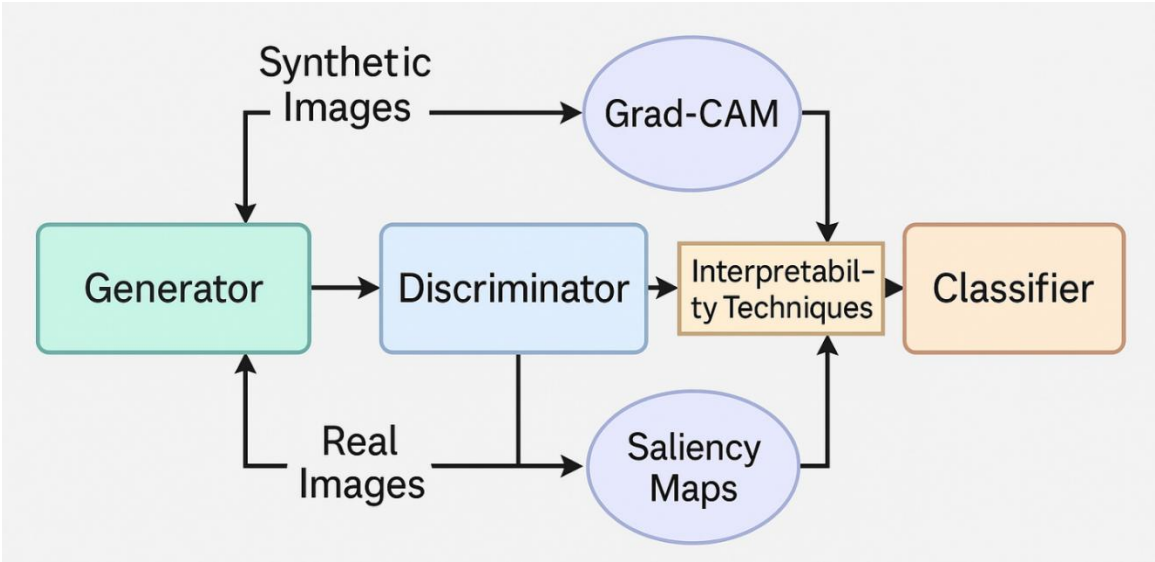

Figure 19 Conceptual Diagram of the XAI Framework Applied

This figure represents Diagram providing a conceptual depiction of the framework employed in this research, which combines discriminator training, quantitative assessment, qualitative visualization (Grad-CAM, Saliency Maps), and interpretation.)

5.3 Practical Implications

5.3.1 Applications in Real-World Scenarios

The enhanced understanding of StyleGAN discriminator behaviour has direct practical applications. In digital content verification, knowledge of the specific cues discriminators use (e.g., textural inconsistencies revealed by saliency maps(Figure 15)) informs the development of more effective deepfake detection algorithms crucial for combating misinformation and fraud [12], [35]. In cybersecurity, understanding discriminator vulnerabilities aids in building systems resilient to adversarial attacks targeting GAN-generated content. For ethical AI, interpretability allows auditing for biases potentially learned from datasets like FFHQ, enabling the implementation of mitigation strategies [13], [16]. In healthcare, interpretable quality control for GAN-generated medical images (used for data augmentation) enhances trust in diagnostic models. Similar benefits apply to privacy-preserving synthetic data generation. On the generative side of GANs, in computer vision tasks, much more is now understood about how to build better generative models [34], [71], or at least what kind of generative models to avoid, e.g., to avoid overfitting, which disallowed prior knowledge from the training set to inform the model.

5.3.2 Implications for GAN Deployment

Discriminator interpretability directly affects usable GAN deployments. It allows for quality control and validation that goes beyond the usual aggregate metrics and enables us to check whether the discriminator is really using meaningful criteria, rather than simply making use of

some unintended dataset bias [35]. This interpretability is not just useful in addressing the common criticism that GANs tend to be black boxes; it also has several practical benefits. For one thing, it allows us to perform better debugging and troubleshooting of the GAN system. When the generator produces a bad output, we can tell from the visualizations what the discriminator is focusing on and thereby get a better handle on what's causing either the generator or discriminator to misbehave [38], [50], [72].

5.3.3 Recommendations for Practitioners

Based on this research, practitioners using StyleGAN or similar models should systematically leverage visualization methods like Grad-CAM and filtered Saliency Maps to comprehend discriminator focus [16]. Saliency maps should be well-filtered or averaged to reduce noise, but results should also be critically assessed for artifacts or misinterpretation [52], [73]. ROC analysis (Figure 12) should be used to optimize thresholds, balancing precision and recall, and confusion matrices (Figure 11) should be referenced to check the kinds of errors being made. These diagnostics should be correlated with visualizations, which should themselves be correlated with any misclassifications that occur, to reason out if high-frequency textures are somehow being generator-faked or just discriminator-sensitized into generating noise [34], [65] Perform a critical evaluation of the training set for potential quality and bias issues, and use interpretability tools to argue for or against the hypothesis that discriminators are learning to rely on some biased feature [35]. Pre-trained models can be used, but only with lots of attention paid to the (expected) adaptation of the discriminator. Finally, keep an eye on the changing landscape of research relevant to XAI and GANs [39].

5.4 Limitations

Several limitations are acknowledged in this study. The reliance on particular gradient-based methods means that the interpretations are subject to the inherent constraints of these methods, which include potential issues with noise, fidelity, and faithfulness [36], [52]. Further, the evaluation metrics (accuracy, precision, recall, F1, and AUC) were calculated on a small sample (N=8) for the final epoch evaluation. While the performance was perfect on this sample, it might not be perfect on a larger, more diverse test set. This finding was based on a specific experimental setup (using one pre-trained generator [8], one specific subset of real data, and certain defined hyperparameters mentioned in Appendix C(8.3)) and might not be universally generalizable to all datasets, all versions of StyleGAN, or all pre-trained models [35]. The chosen pre-trained generator makes a difference in interpretation [56]. Finally, GAN interpretability is an emerging field, and the rapid pace of development might make some results obsolete in a very short timeframe [16]. The limitations just discussed frame the scope of this study's conclusions.

Table 7 Summary of Study Limitations

Limitation	Description
Gradient Method Limits	Saliency artifacts/noise possible
Evaluation Sample Size	Perfect score on small sample (N=8)
Dataset/Model Specificity	Used specific FFHQ subset & StyleGAN version
Fixed Generator Context	Interpretations tied to one generator
Computational Constraints	Limited scale of hyperparameter search/trials
Interpretability Metrics	Lack of quantitative faithfulness metrics
Architectural Constraints	Standard CNN discriminator used

6 Conclusion and Future Work

6.1.1 Summary of Findings

This dissertation presented a detailed study of the interpretability of discriminators in StyleGANs. The main aim was to probe into the decision-making of a discriminator trained using the StyleGAN framework against high-quality pretrained generators. We set out to answer two main questions throughout the course of this study. The first one was a simple yet fundamental question: What features does a StyleGAN discriminator use to make decisions, i.e. classifying images as real or fake? The second question sought to gain deeper insights into the first: Why does a StyleGAN discriminator use those features to make decisions?

Chapter 4 lays out the results in detail. They show that the trained discriminator really works. It distinguishes real images from synthetic ones, and it does so not just based on some obvious gross differences ('high-level semantic features' like the eyes, nose, and mouth) but also, and most crucially, based on some subtle, texture-related, difference-making details ('fine-grained textural details'). While the classifier works on this small sample of N=8 images, achieving perfect performance at this small sample size is a little worrisome because of the overfitting potential it implies.

6.1.2 Contributions

This research contributes to the fields of GANs and XAI in several ways. First, it applied standard XAI methods (Grad-CAM and filtered Saliency Maps) to interpret the discriminator of a state-of-the-art generative model (StyleGAN) for the first time and validated their use in this domain. Second, the research provided empirical evidence regarding the learned behaviour of the StyleGAN discriminator, which was shown to differentiate real from fake images using a combination of semantic structures (the kind of faces that are seen) and fine-grained textural details (really small textures that you can see if you look closely) along with the use of a pretty high- level kind of interpretability that XAI really aims for and which is not obtained with off-the-shelf image classifiers. Third, we presented a holistic methodological framework for combined performance evaluation of kind of a quantitative (evaluation metrics) and a qualitative kind of interpretability analysis (the use of XAI methods), and we argued for why a more comprehensive format for just reporting the use of evaluation protocols should replace the just reporting of FID scores [68]. If GANs are to be used in for important use cases, they need to be made more understandable and transparent [12].

6.1.3 Limitations

As analysed thoroughly in Chapter 5 (particularly in Section 5.4 Table 7), this work accepts certain restrictions. It is inescapable that gradient-based methods, like those used here, have aspects that are not completely understood. This study interprets the results from the specific techniques of Grad-CAM and Saliency Maps, both of which have the potential to obscure what the model is doing. Even when these techniques do not produce obvious artifacts, they rely on a fundamentally flawed understanding of the relationship between gradients and the structure of the learned representation; indeed, one might argue that there are significant aspects of neural networks and computer vision that are understood poorly and are represented poorly in

these techniques [35], [61]. In cases when Grad-CAM or Saliency Maps are interpreted, one can certainly question mechanistically what the model is doing. These are essentially the limits and problems with the methods and interpretations of this work.

6.2 Future Work

This thesis's findings and limitations suggest several fruitful avenues for future work. One main direction concern improving visualization methods. We could investigate alternative or complementary interpretability methods beyond Grad-CAM and the filtered Saliency map, like Integrated Gradients [43] or Smooth Grad [74]. These may result in a more robust and potentially more insightful understanding of how our discriminator works. Likewise, we could consider mechanisms based on attention. Either analysing them when they're present or using them in our discriminator architectures may yield more interpretable models.

Another important line of work is assessing generalizability and robustness. It's vital to put the trained discriminator and the generated interpretation results to the test on a wider array of datasets beyond just the FFHQ subset - like CelebA, MetFaces, or other non-face datasets - and also with different versions of StyleGAN and other GAN architectures [9], [35] .Not to mention, it's super important to see how well the interpretations hold up under pressures from attacks designed to make explanation methods look bad. Even more crucially, we feel it's necessary to assess the robustness of the interpretations themselves against similar adversarial attacks that are supposed to fool the explanation methods.

It is also necessary to develop more advanced evaluation metrics. This entails constructing numerical metrics that are tailored to gauge the fidelity and dependability of the explanations that are produced for GAN discriminators. Visual inspection and standard classification metrics are not sufficient.

Additional inquiries might also explore the theoretical foundations of the learned decision boundary in greater depth, possibly with the aid of information theory or causal analysis, to comprehend why certain features are preferred over others. Our understanding of the relationship between discriminator interpretability and GAN training dynamics-such as mode collapse or convergence stability- could be greatly enriched by exploring this link further. Likewise, probing the reasons for our observed perfect sample performance-whether due to specific artifact detection, overfitting, or limits in our evaluation framework-seems a worthwhile and potentially illuminating endeavour.

6.3 Concluding Remarks

This thesis offered an intensive overview of the interpretability of StyleGAN discriminators. While it is very important for high-stakes models to be interpretable, we have not been applying the detailed analysis of XAI in complex yet state-of-the-art discriminators like those of

StyleGAN. The insights from this research are relevant to the much-studied performance validation of these models, their never-discussed model robustness, and the crucial characteristic of transparency in these complex decision-making constructs. High-stakes models that offer poor interpretability can hardly be regarded as responsible even when they are deployable. Showing how XAI tools can be applied to a StyleGAN decision-making device, this work offers insights into how and why that device makes decisions based on both semantic and textural features. In short, this work contributes to the understanding of these devices as we build more robust, trustworthy models. And generative models that make complex decisions can hardly be regarded as interpretable if their decision-making devices are opaque.

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8 Appendices

8.1 Appendix A: Exploratory Experiments and Intermediate Results

Purpose: To document the iterative process, earlier trials, or different analysis approaches explored during development.

Figure 20 Example GradCAM and Saliency Map visualization

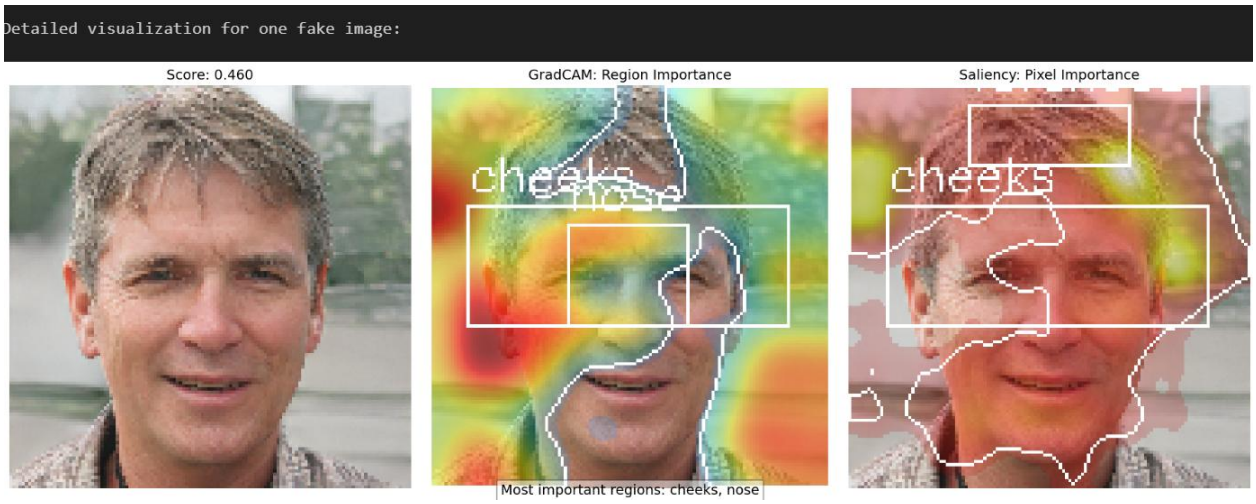

Figure 21 Detailed Grad CAM and Saliency visualization for a sample real image during exploratory analysis, highlighting specific feature regions.

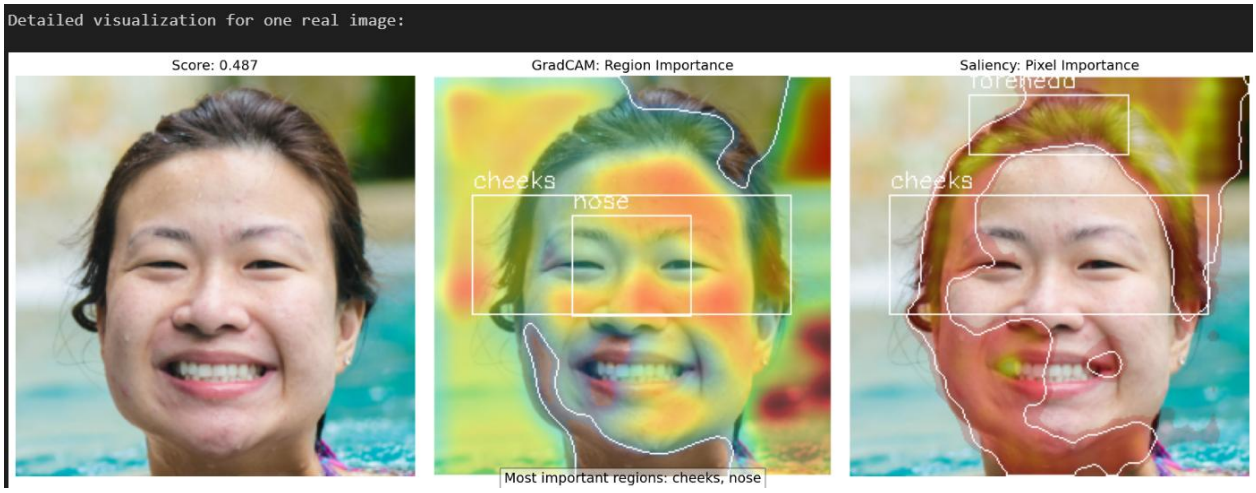

Figure 22 Detailed Grad CAM and Saliency visualization for a sample fake image during exploratory analysis

8.2 Appendix B: Sample Outputs from Pre-trained StyleGAN Generator

Purpose: To provide visual context for the synthetic images used as the 'fake' class.

Figure 23 Representative sample images generated using the pre-trained StyleGAN (FFHQ 1024x1024) model

8.3 Appendix C: Final Training Hyperparameters and Configuration

Purpose: To provide precise details for reproducibility of the final successful training run.

Table 8 Final Hyperparameters for Discriminator Training

Hyperparameter	Value	Description
LEARNING_RATE	2e-5	Learning rate used in AdamW optimizer
BETAS	(0.5, 0.99)	Beta coefficients for AdamW (momentum terms)
WEIGHT_DECAY	5e-5	Regularization term to prevent overfitting
DROPOUT_RATE	0.3	Dropout applied in discriminator convolutional layers
INPUT_NOISE_STD	0.03	Standard deviation for Gaussian noise added to real/fake input images
LABEL_SMOOTHING_REAL	0.9	Soft label for real images to improve generalization
LABEL_SMOOTHING_FAKE	0.1	Soft label for fake images
BATCH_SIZE	16	Number of real images per batch during training
NUM_EPOCHS	100	Total training iterations over the dataset
OPTIMIZER	AdamW	Optimizer used for training the discriminator
LOSS_FUNCTION	Binary Cross-Entropy (BCE)	Used to compute classification error between predicted and true labels
NORMALIZATION	InstanceNorm2D	Normalization strategy inside discriminator blocks
AUGMENTATION	Gaussian Noise Injection	Applied to both real and fake images
EVAL_FREQUENCY	Every 10 epochs	Evaluation (Grad-CAM + Saliency) every 10 epochs
REAL_DATASET	FFHQ subset (real/class0 dir.)	High-quality facial dataset for real samples
GENERATOR	Pretrained StyleGAN (FFHQ 1024)	Used as a fixed generator during discriminator training

8.4 Appendix D: Supplementary Visualizations

Purpose: To include additional visualizations using the final model/methodology that supplement the main results chapter.

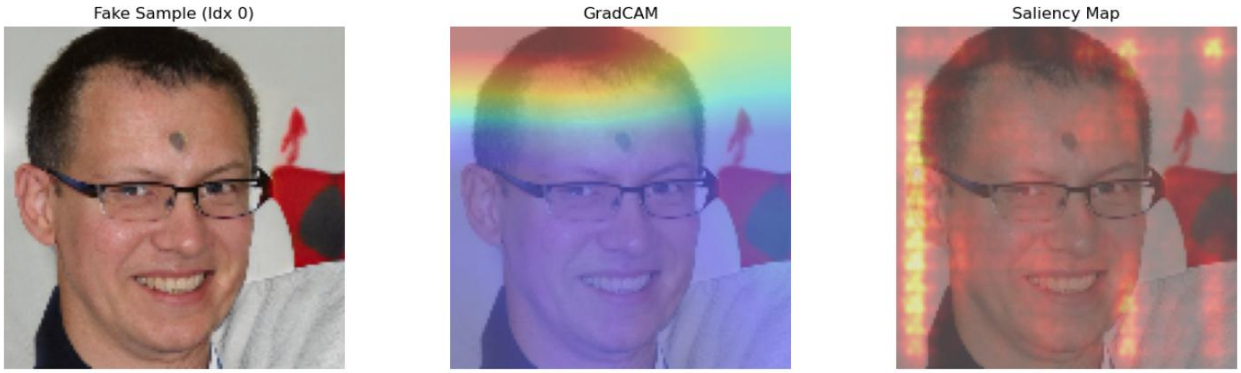

Figure 24 Additional example of Grad CAM and Saliency Map visualization for a fake sample using the final model.

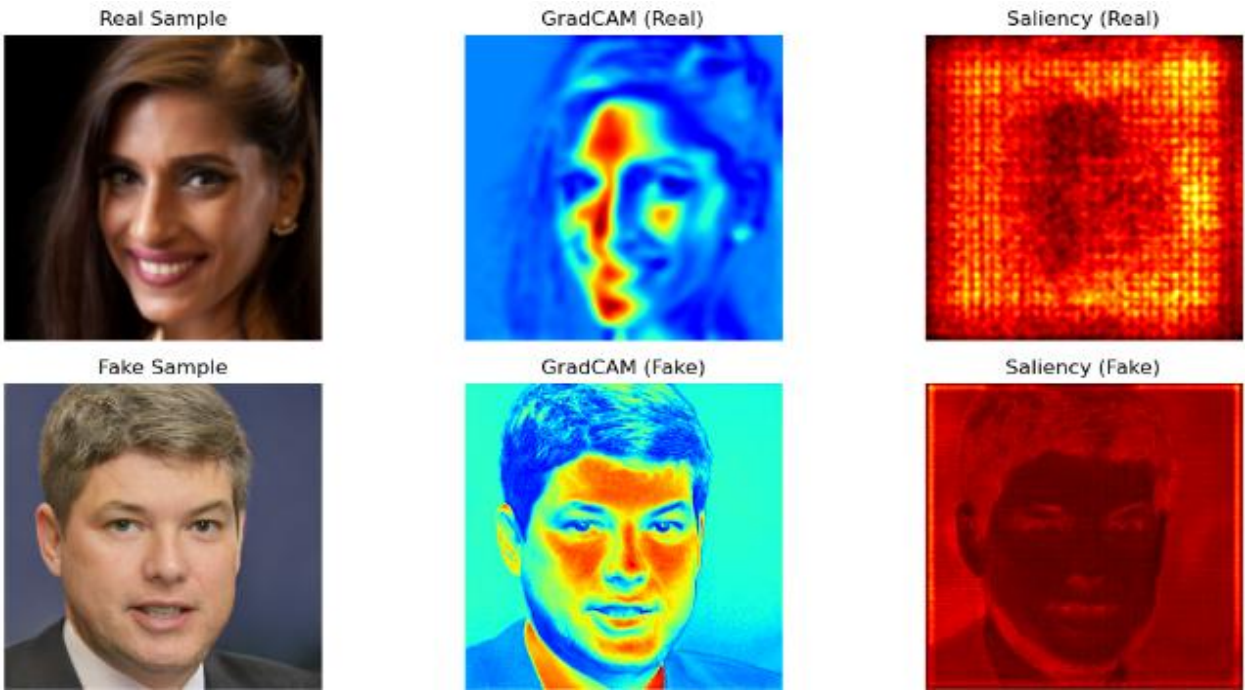

Figure 25 Pairwise comparison of Grad CAM and Saliency visualizations for a sample Real image (Top Row) and Fake image (Bottom Row)

8.5 Appendix E: Computational Environment

Purpose: To document the specific hardware and software environment used.

Table 9 Hardware and software specifications

Component	Specification
GPU	NVIDIA GeForce RTX 4090 (24 GB VRAM, CUDA Compute Capable)
GPU Driver	Version 550.120
CUDA Version	12.4
RAM	Please confirm total system memory (e.g., 64 GB DDR4)
Operating System	Ubuntu 22.04 LTS / GNOME Desktop (inferred from process list)
Python Version	3.11 (please confirm if different)
PyTorch Version	2.5.1+ py3.11_cuda12.4_cudnn9.1.0_0
CUDA Toolkit	12.4 (aligned with PyTorch compatibility)
X Server Processes	Xorg & GNOME Shell

8.6 Appendix F: Accessing Project Files

All the project files, including hardware design, custom IPs, and software for the projects, are uploaded to a GitLab (OTH Regensburg) repository. This repository is open to use by everyone as a reference for this thesis.

The full project, including all source code, design files, and detailed instructions for setting up and running the project, can be accessed at:

https://gitlab.oth-regensburg.de/EI/Labore/ses-labor/interpreting-deep-neural-networks-explaining-the-methods-behind-ai-generated-image-recognition